

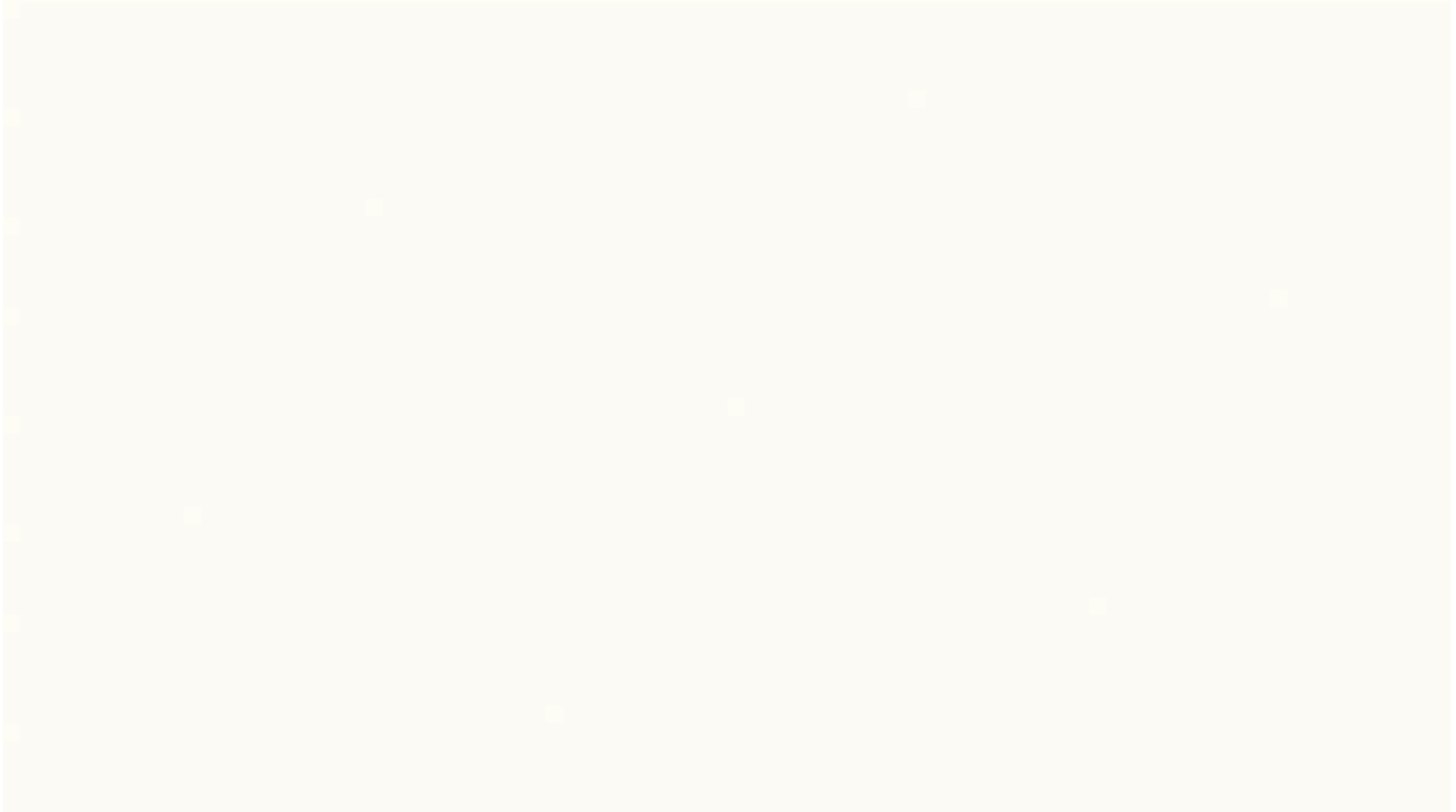


Eötvös Loránd University
Faculty of Informatics

UX for the web

Web-development I. course

What is UX?



<https://www.youtube.com/watch?v=bsla797t2Vk>



The Fundamentals of UX

What is UX?

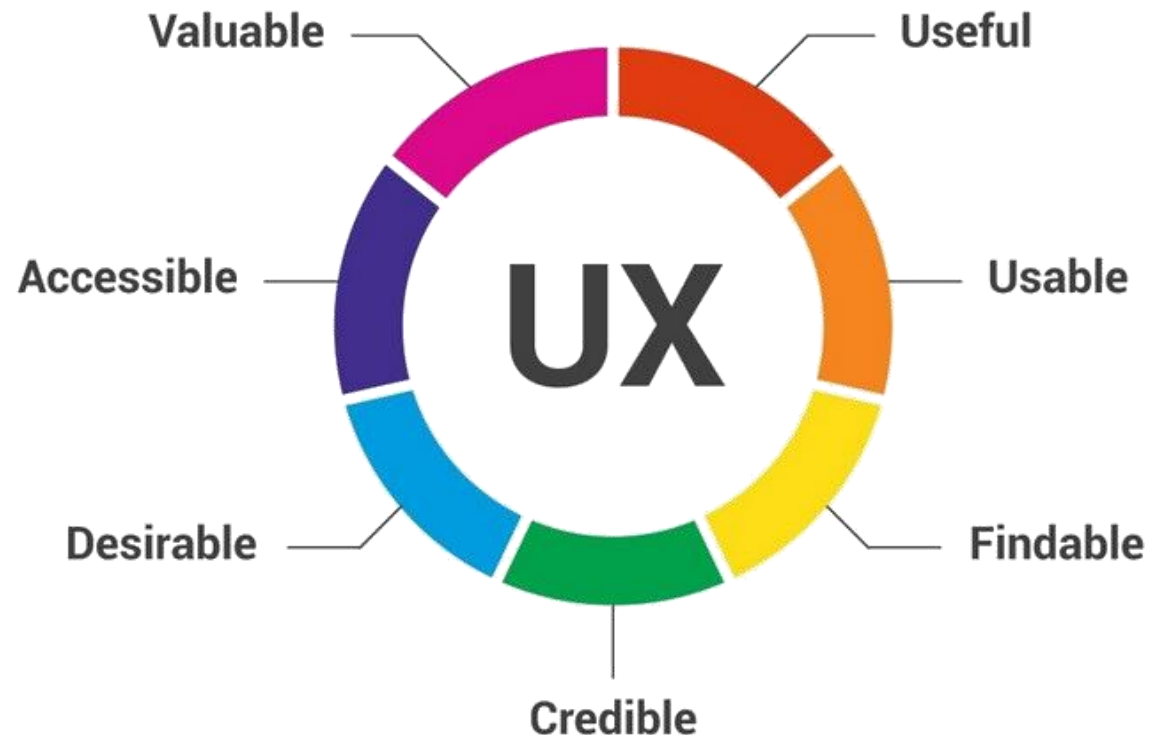
“‘User experience’ encompasses all aspects of the end-user’s interaction with the company, its services, and its products.”

The term UX was coined by cognitive scientist Don Norman in the early 1990’s while he was VP of the Advanced Technology Group at Apple.



User experience (UX)

UX is a broad discipline with an intricate set of interlinked components that contribute to the overall experience a user will have with a digital product, however, the core focus is usability.



7 factors

1. Useful

- If a product isn't useful to someone, why would you want to bring it to market?

- <http://www.theuselessweb.com/>



7 factors

2. Usable

- Usability is concerned with enabling users to achieve their end objective with a product effectively and efficiently.

The Amazon logo, featuring the word "amazon" in a bold, black, sans-serif font. Below the text is a curved orange arrow that starts under the 'a' and points towards the 'n'.The eBay logo, featuring the word "ebay" in a lowercase, sans-serif font. Each letter is a different color: 'e' is red, 'b' is blue, 'a' is yellow, and 'y' is green.The Netflix logo, featuring the word "NETFLIX" in a bold, red, sans-serif font. The letters are slightly shadowed, giving it a 3D appearance.

Useful vs usable

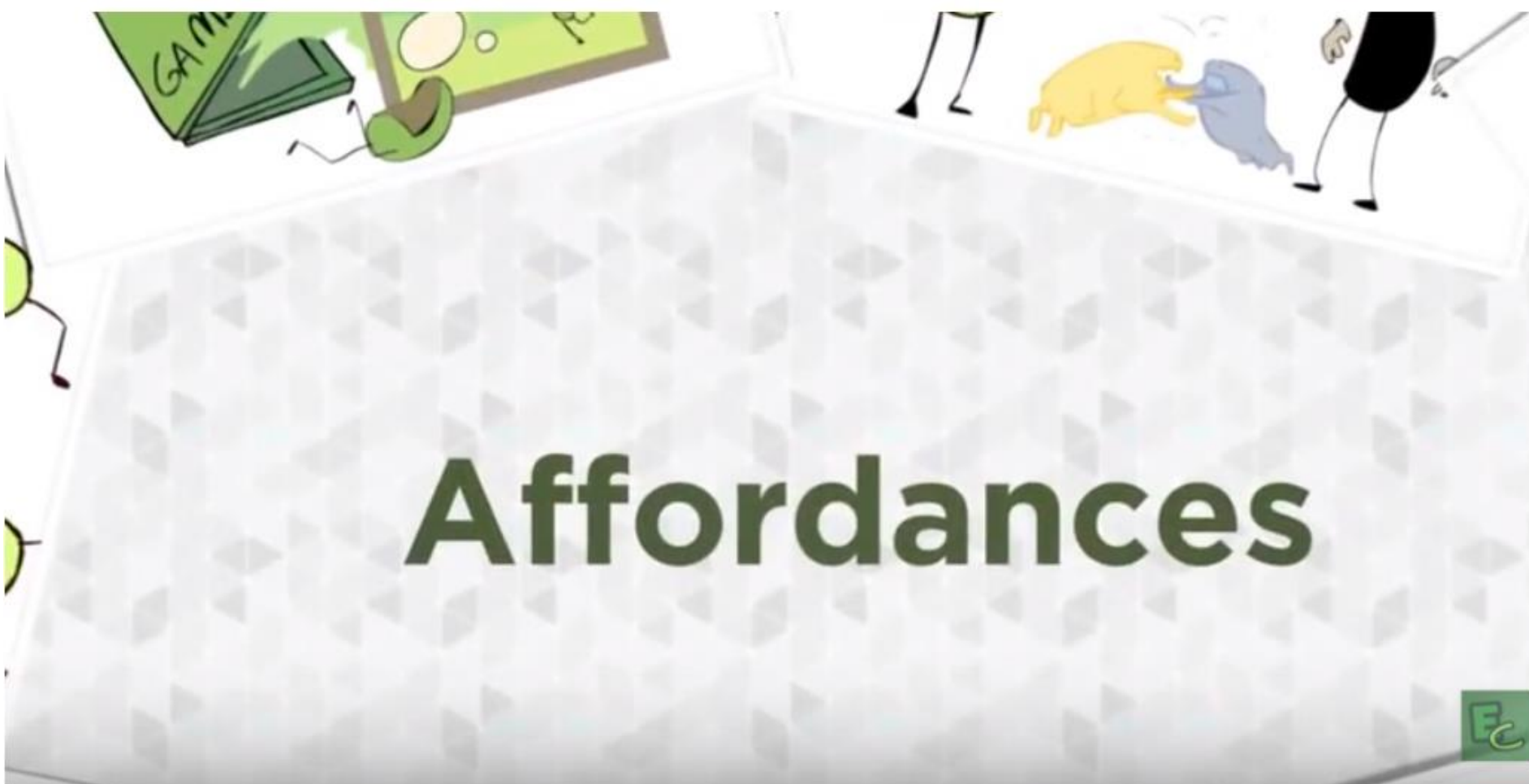
- Many “useful” products fail to be “usable”



Useful vs usable

- A good product needs to engage its users and provide enough affordance (a quality of an object, or an environment, which allows an individual to perform an action) to provide a good user experience or complete the intended action for the user.

Affordances - How Design Teaches Us Without Words



Affordances

<https://www.youtube.com/watch?v=QCSXEKHL6fc>

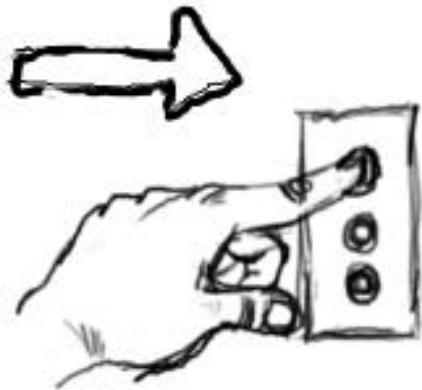
Useful vs usable



<https://50daysux.wordpress.com/tag/usability/>

Affordances

- Affordances are an object's properties that show the possible actions users can take with it, thereby suggesting *how* they may interact with that object.



Button - Push



Switch - Flip



Knob - Rotate





If it needs help text
it's bad design.

Can't they just follow
standard conventions?

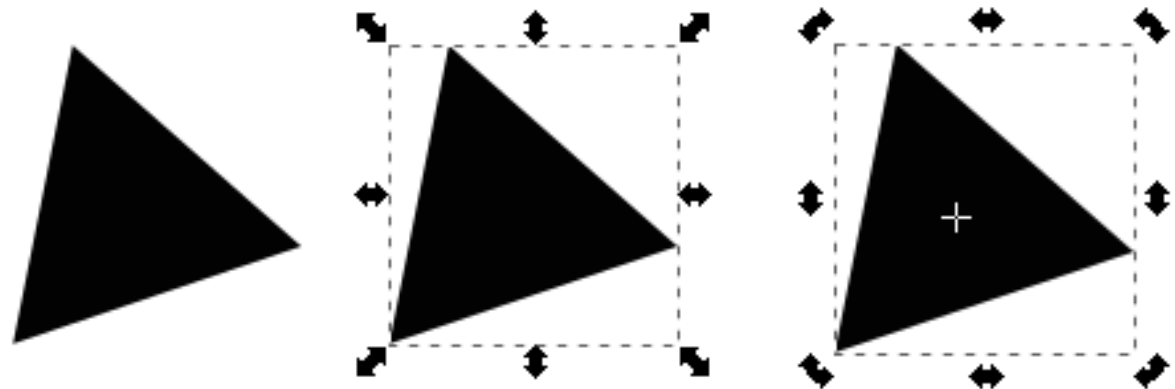


USABILITY PEOPLE IN A RESTROOM.

S.K.
2013

Affordances

- To create a good interface, must create it with appropriate affordances to make user's task easy.
- E.g. when object selected, make handles appear that object might be moved (or something)



Affordances

Buy now

Buy now

- A button „invites” pushing
- A door handle „affords” pullings

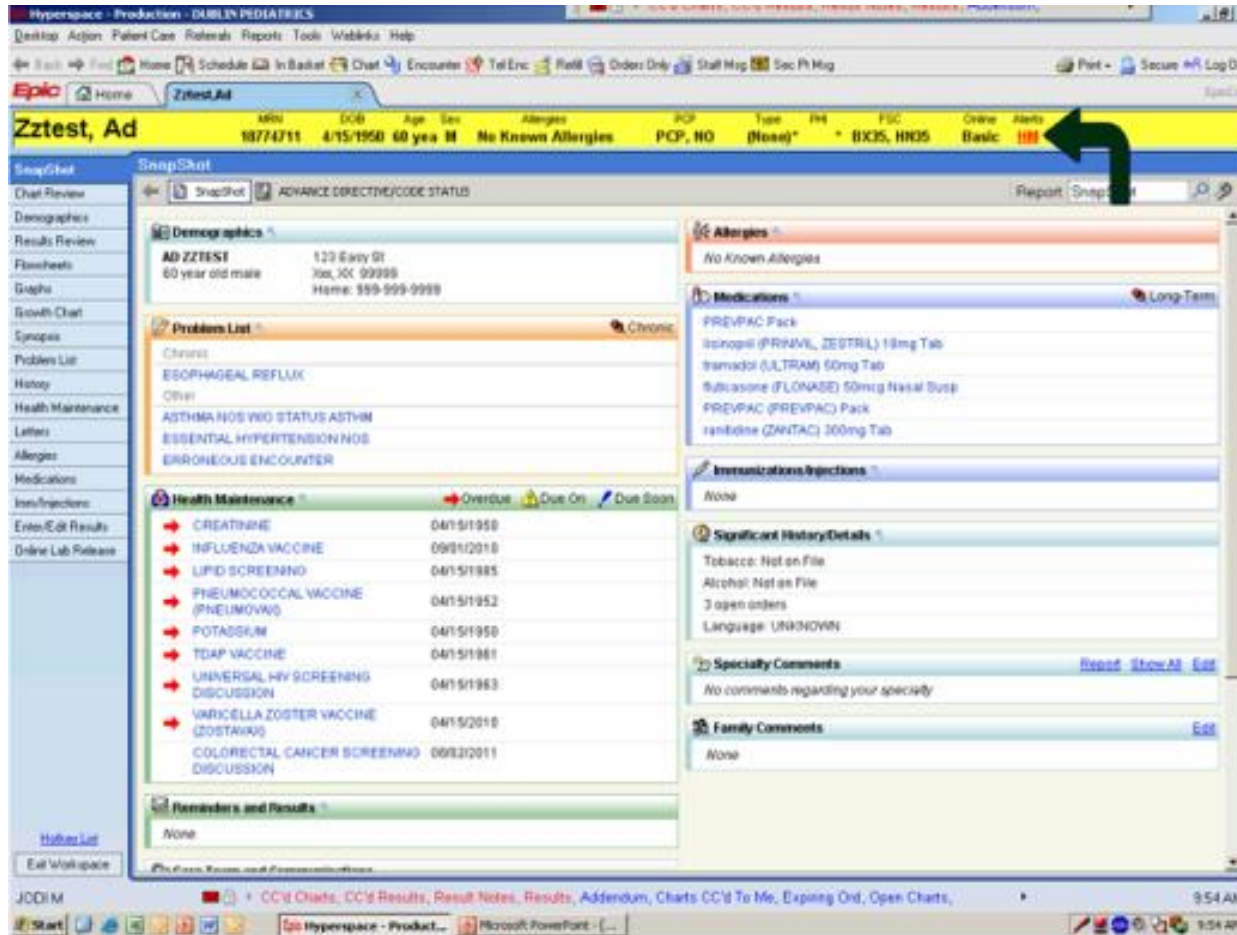


7 factors

3. Findable

- Findable refers to the idea that the product must be easy to find, and in the instance of digital and information products, the content within them must be easy to find, too.

Overly complex interfaces



Software needs to be simple and intuitive in order for people to accomplish their given tasks with minimum difficulty

7 factors

4. Credible

- Twenty-first-century users aren't going to give you a second chance to fool them—there are plenty of alternatives in nearly every field for them to choose a credible product provider.
- They can and will leave in a matter of seconds and clicks unless you give them reason to stay.

Deceptive Imagery

- Are you ready for your tropical vacation? Err... I mean, haircut.



Users react to visuals faster than text, so always make sure your content matches the supporting visuals.

7 factors

5. Desirable

Desirability is conveyed in design through branding, image, identity, aesthetics, and emotional design. The more desirable a product is, the more likely it is that the user who has it will brag about it and create desire in other users.



7 factors

6. Accessible

- Accessibility is about providing an experience which can be accessed by users with a full range of abilities—this includes those who are disabled in some respect, such as the hearing, vision, motion, or learning impaired



7 factors

7. Valuable

- The product must deliver value. It must deliver value to the business which creates it and to the user who buys or uses it. Without value, it is likely that any initial success of a product will eventually corrode as the realities of natural economics start to undermine it.

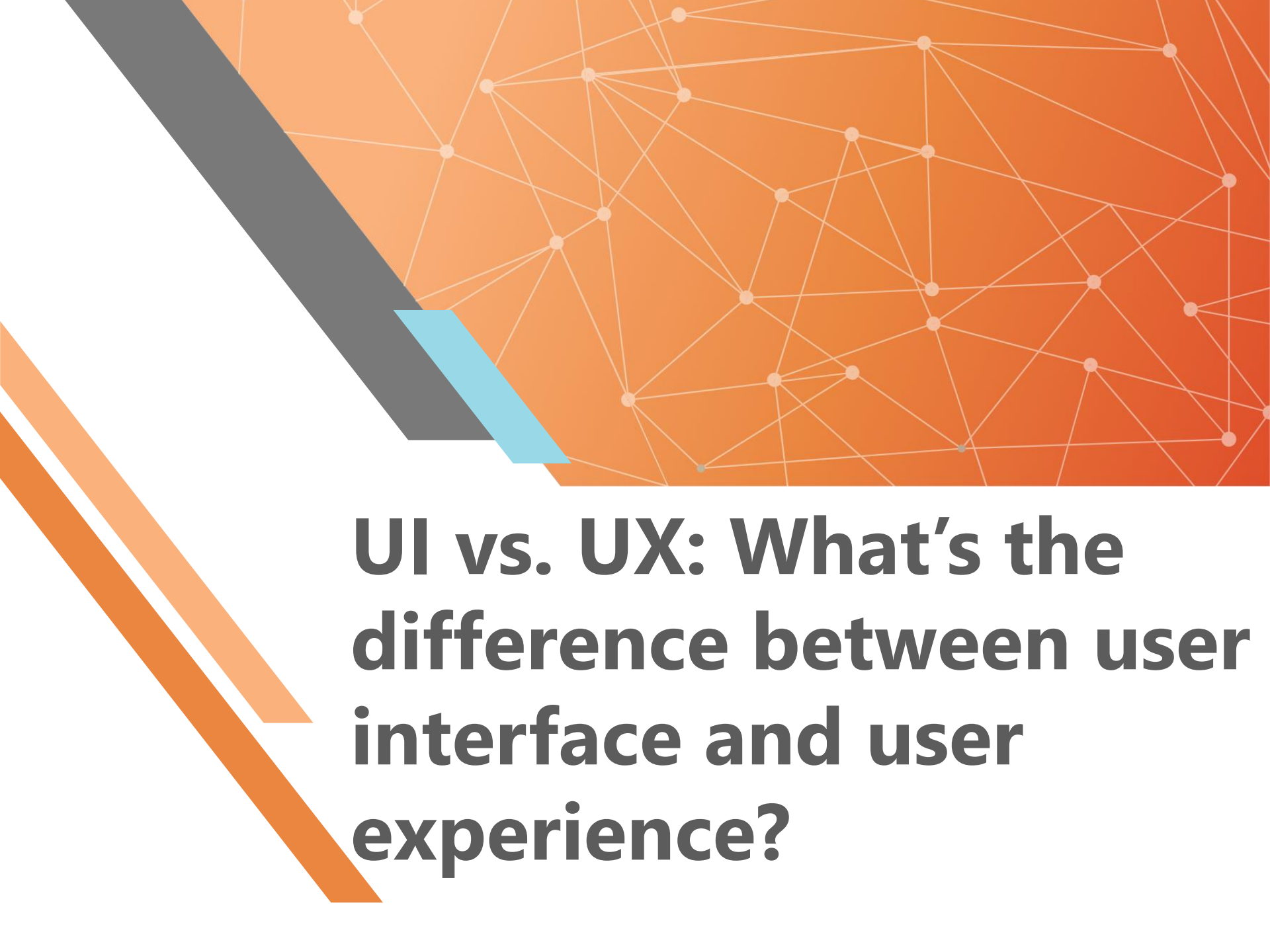


II. What is User Experience (UX)?



Defining this Alphabet Soup!

- Most UX professionals use these terms interchangeable - and usability and user experience (UX) often mean the same things
- Often these terms have distinct and different definitions between UX professionals
- Many people are adding UX to their resume without an solid understanding of the profession as well and understanding the development of one's UX skill set is critical



UI vs. UX: What's the difference between user interface and user experience?

UI vs UX

- At the most basic level, the **user interface** (UI) is the series of screens, pages, and visual elements—like buttons and icons—that you use to interact with a device.
- **User experience** (UX), on the other hand, is the internal experience that a person has as they interact with every aspect of a company's products and services.

UX is focused on the user's journey to solve a problem, UI is focused on how a product's surfaces look and function

UI vs UX

“It’s important to distinguish the total user experience from the user interface (UI), even though the UI is obviously an extremely important part of the design.

As an example, consider a website with movie reviews. Even if the UI for finding a film is perfect, the UX will be poor for a user who wants information about a small independent release if the underlying database only contains movies from the major studios.”

Don Norman and Jakob Nielsen



UX fails

<https://thenextweb.com/dd/2015/09/29/6-examples-of-awful-ux-design/>

Overly Complex Password Protections

Password cannot contain any dictionary words and it must meet the following criterias:

- ✓ Password must be 12 characters long
- ✓ At least 1 character must be alphabetical AND at least 1 character must be a digit OR a special character: ! # \$ %
- ✓ At least 6 characters must occur only once in a password
- ✓ Passwords cannot contain any string that is also contained in the username
- ✓ Passwords cannot contain any common strings such as a sequential series of letters (abcd) or a sequential series of numbers (1234) or pattern of numbers (2468)

Old Password:

New Password:



Confirm New Password:



Continue

Cancel

Long Forms with Quick Resets

* - Denotes Required Information

> 1 Donation > 2 Confirmation > Thank You!

Donor Information

First Name*

Last Name*

Company

Address 1*

Address 2

City*

State*

Zip Code*

Country*

Phone

Fax

Email*

Donation Amount* ☒ None ☐ \$50 ☐ \$75 ☐ \$100 ☐ \$250 ☐ Other
(Check a button or type in your amount) Other Amount \$

Recurring Donation ☐ I am interested in giving on a regular basis.
(Check if yes) Monthly Credit Card \$ For Months

Honorarium and Memorial Donation Information

I would like to make this donation ☐ To Honor ☐ In Memory of

Name

Acknowledge Donation to

Address

City

State

Zip

Additional Information

Please enter your name, company or organization as you would like it to appear in our publications.

Name

☐ I would like my gift to remain anonymous.
☐ My employer offers a matching gift program. I will mail the matching gift form.
☐ Please save the cost of acknowledging this gift by not mailing a thank you letter.

Comments
(Please type any questions or feedback here)

How may we contact you? ☐ E-mail ☐ Postal Mail ☐ Telephone ☐ Fax

I would like to receive newsletters and information about special events by:
☐ E-mail ☐ Postal Mail

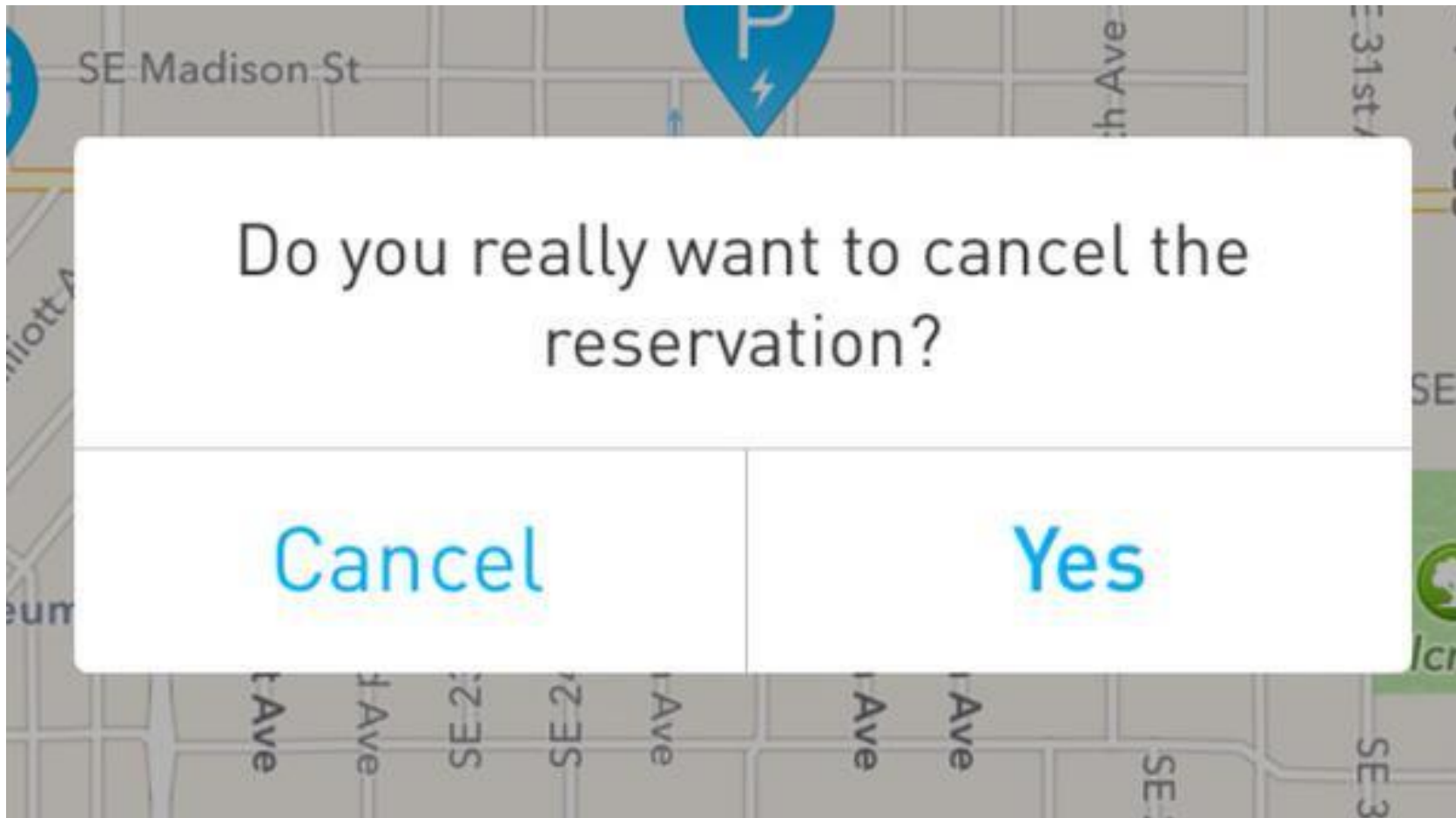
☐ I would like information about volunteering with the

Donate online with confidence. You are on a secure server.
 If you have any problems or questions, please contact support.



Users can accidentally erase everything once they've finished

Epic fails





Usability

What is Usability?

- Ease of learning
- Ease of use
- Ease of remembering
- Subjective satisfaction
- Efficiency of use
- Effectiveness of use





International Organization for Standardization (ISO) - Usability

"[Usability refers to] the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use." - ISO 9241-11

Usability

- **Effectiveness** is the accuracy and completeness with which specified users can achieve specified goals in particular environments.
- **Efficiency** is the resources expended in relation to the accuracy and completeness of the goals achieved.
- **Satisfaction** is the comfort and acceptability of the work system to its users and other people affected by its use.

Usability is defined by 5 quality components:

U S A B I L I T Y

Learnability: How easy is it for users to accomplish basic tasks the first time they encounter the design?

Efficiency: Once users have learned the design, how quickly can they perform tasks?

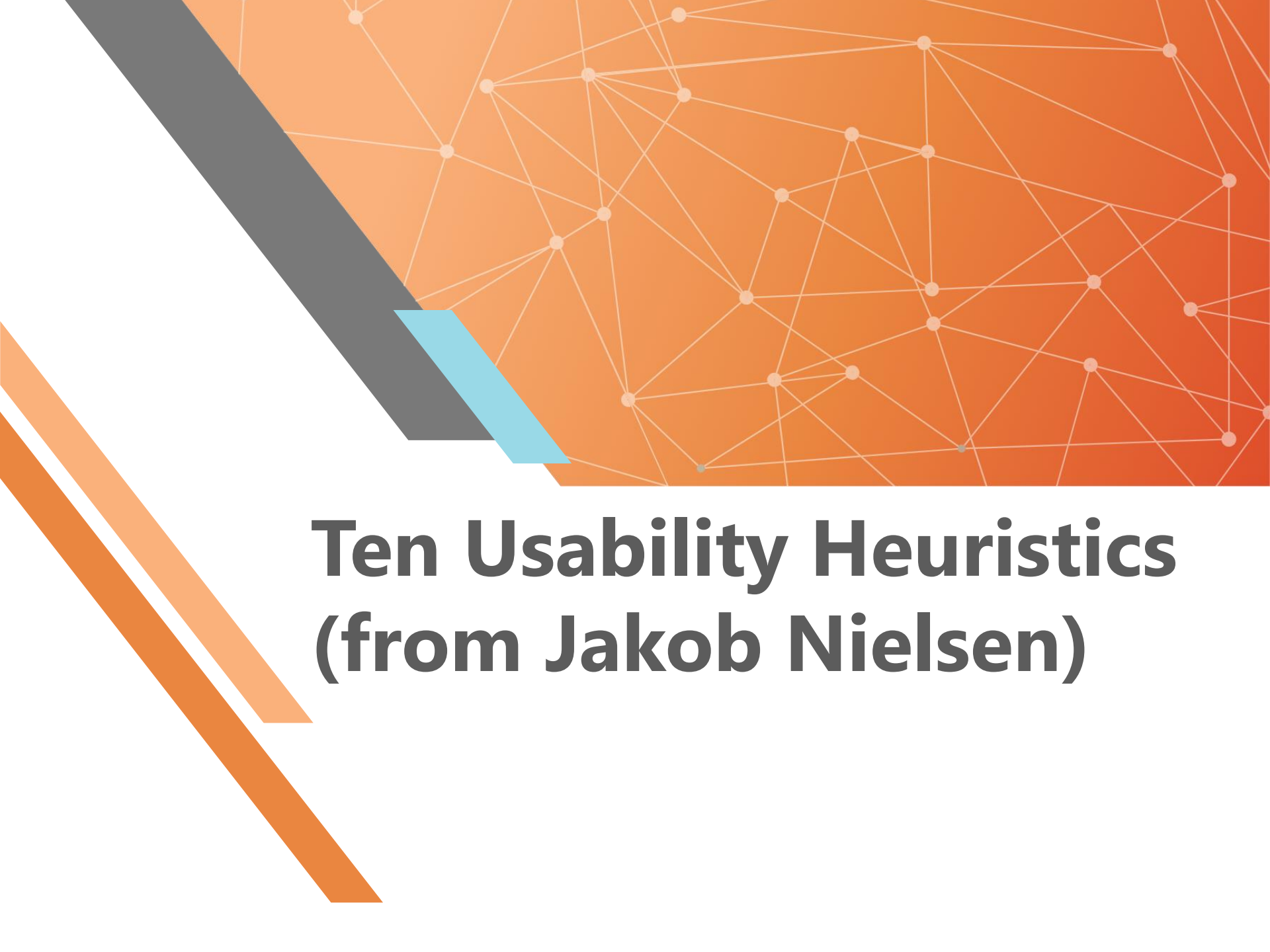
Memorability: When users return to the design after a period of not using it, how easily can they reestablish proficiency?

Errors: How many errors do users make, how severe are these errors, and how easily can they recover from the errors?

Satisfaction: How pleasant is it to use the design?

Utility, Usability

- Definition of **Utility** = whether it provides the **features you need**.
- Definition of **Usability** = how **easy & pleasant** these features are to use.
- Definition of **Useful** = **usability + utility**.

The background features a complex geometric design. On the left, there are several diagonal stripes in shades of orange and grey. On the right, a network diagram is visible, consisting of numerous small white dots connected by thin white lines, all set against a gradient of orange and red. The title text is positioned in the lower right area, overlapping the white background.

Ten Usability Heuristics (from Jakob Nielsen)

Usability Heuristics

- Avoid common design pitfalls by following principles of good design
- Jakob Nielsen proposes 10 heuristics, others propose more or less.
- Inspect an interface for usability problems with these principles



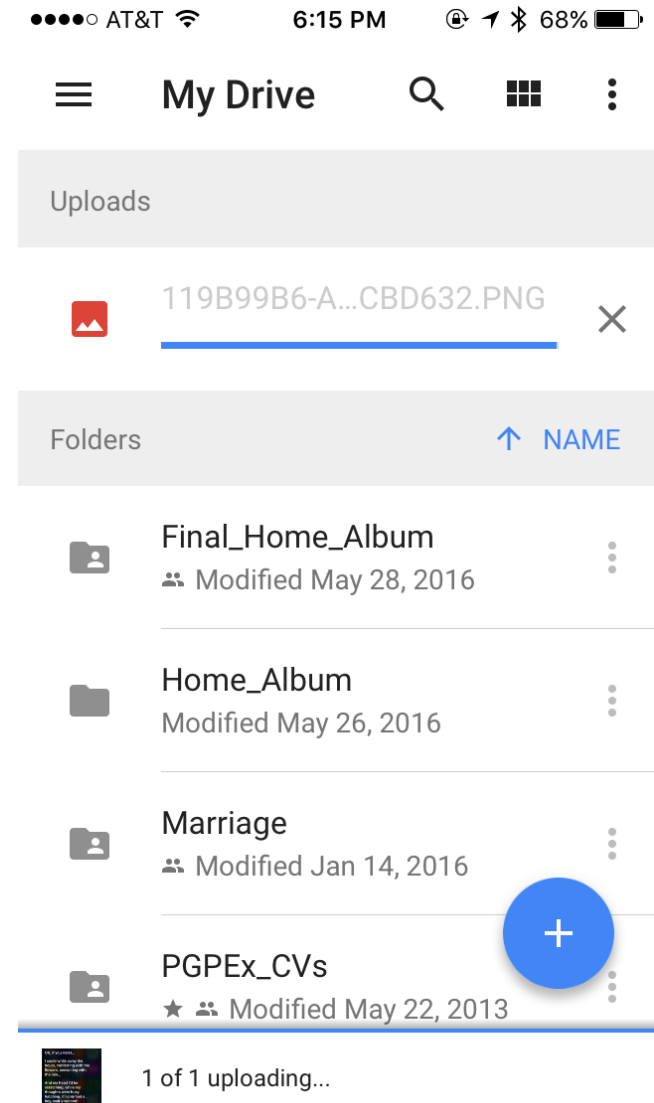
Jakob Nielsen, Ph.D., is a User Advocate and principal of the Nielsen Norman Group

"the guru of Web page usability"
(The New York Times)

1. Visibility of system status

The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.

Google Drive showing the status of a document upload



1. Visibility of system status

Like how Twitter tells you when it's publishing a tweet and makes a chirp sound when it's done

United Airlines Flight 410		
On-time - departs in 1 hour 41 mins		
PDX		DEN
Departs Portland, today		
Time	Terminal	Gate
11:48 AM	-	E2
Arrives Denver, today		
Scheduled 3:18 PM	Terminal	Gate
3:17 PM	-	B27

United Airlines Flight 410		
Delayed (21 mins) - arrives in 57 mins		
PDX		DEN
Departs Portland, today		
Scheduled 11:48 AM	Terminal	Gate
11:49 AM	-	E2
Arrives Denver, today		
Scheduled 3:18 PM	Terminal	Gate
3:39 PM	-	B27



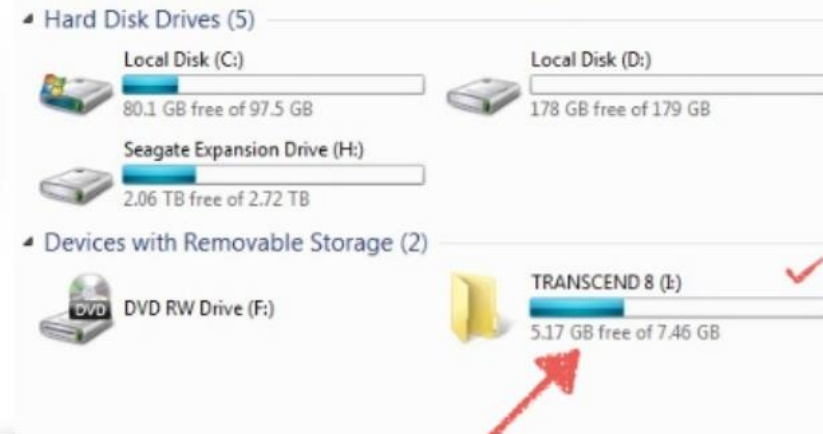
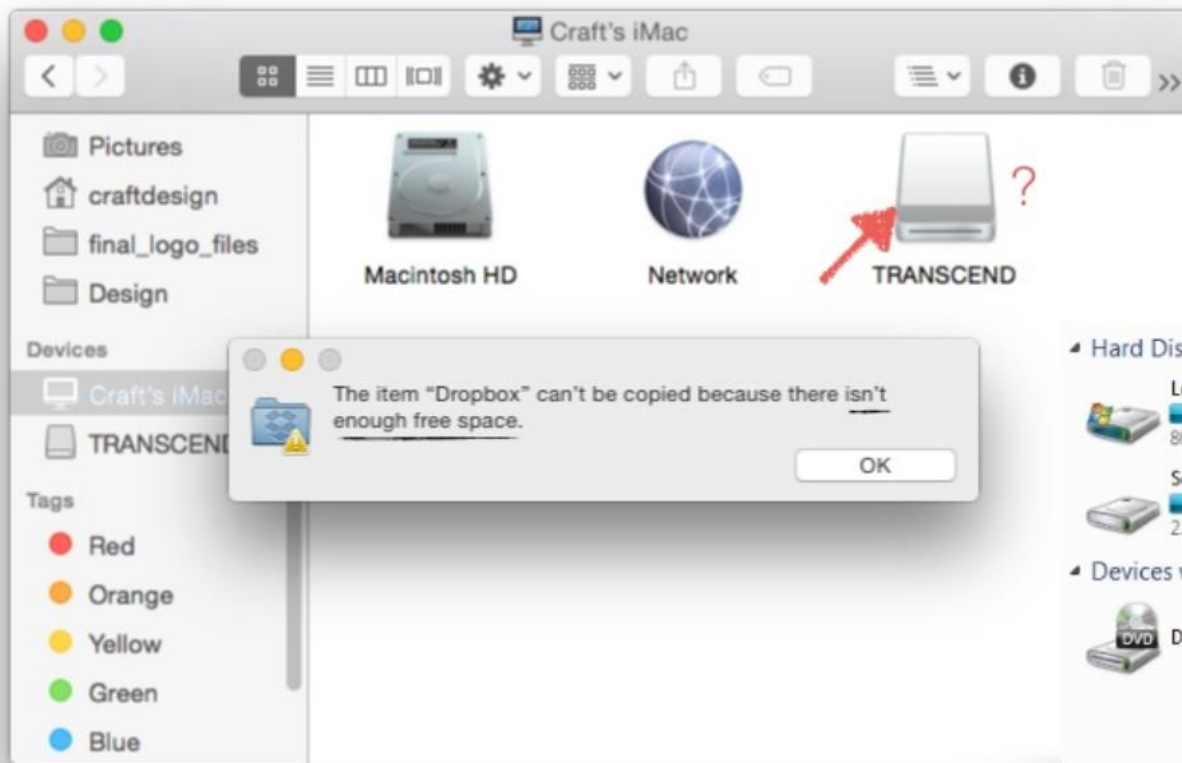
Or how the colour and position of the plane changes on Google search results to indicate if a flight is on time

1. Visibility of system status



Don't keep your users guessing.

Would it be better if the system told you how much space you have on your drive before you begin a file transfer?



2. Match between system and the real world

The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system-oriented terms. Follow real-world conventions, making information appear in a natural and logical order.

2. Match between system and the real world

Buttons can be like real world conversations instead of labels.



Clearer and more effective than "FAQs"

How can we help you?

Search help

Live Updates

Delayed Response from Customer Care

We're really busy at the moment which means it may take a [little](#) longer than usual to get back to you. [More Info](#)



ORDERS

- Amend order
- Cancel order
- Sizing and care guides

[SEE ALL](#)



PAYMENT

- Payment types
- Promo codes
- When will I be charged?

[SEE ALL](#)



DELIVERY

- Where's my order?
- International deliveries
- UK deliveries

[SEE ALL](#)



RETURN

- Have you
- Have you
- Can I ex

2. Match between system and the real world



Refrain from confusing people with system oriented language and design.

oops, there is a problem

Target.com requires **cookie** to be enabled.



continue if enabled

3. User control and freedom

Users often choose system functions by mistake and will need a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue. Support undo and redo.

3. User control and freedom

Allow people to change or delete items in a shopping cart as seen in this example from Inkling. It's also useful to allow them to continue shopping.

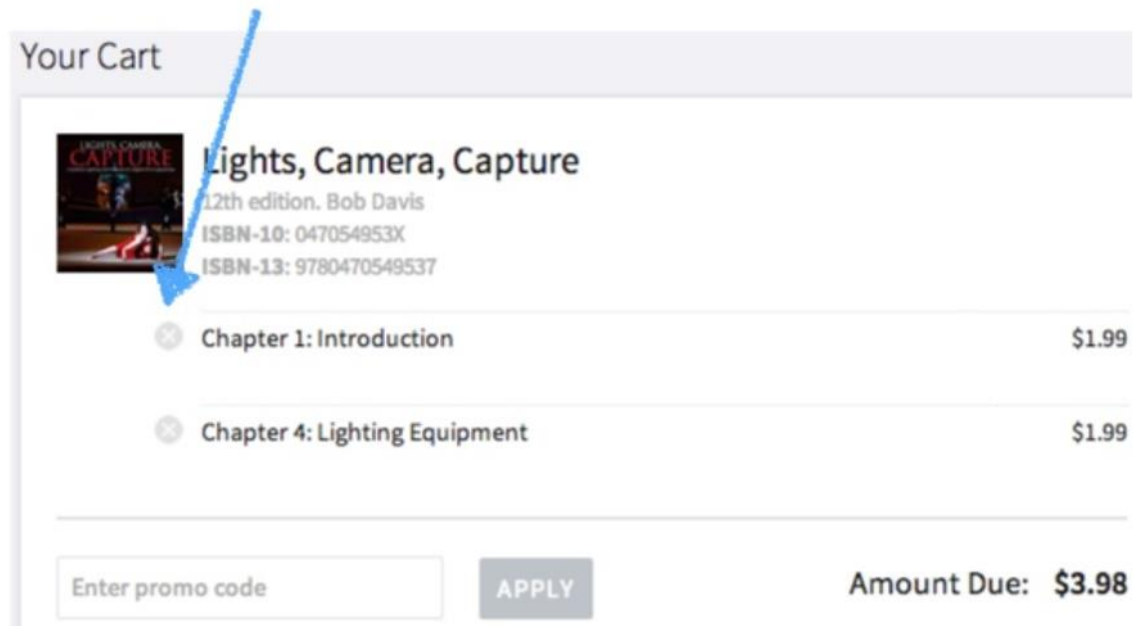
Sending... [Cancel](#)

Your message has been sent. [Undo](#) [View message](#)



Ever felt the need for an undo button after sending an email to the wrong person?

It's a good thing Gmail allows that.

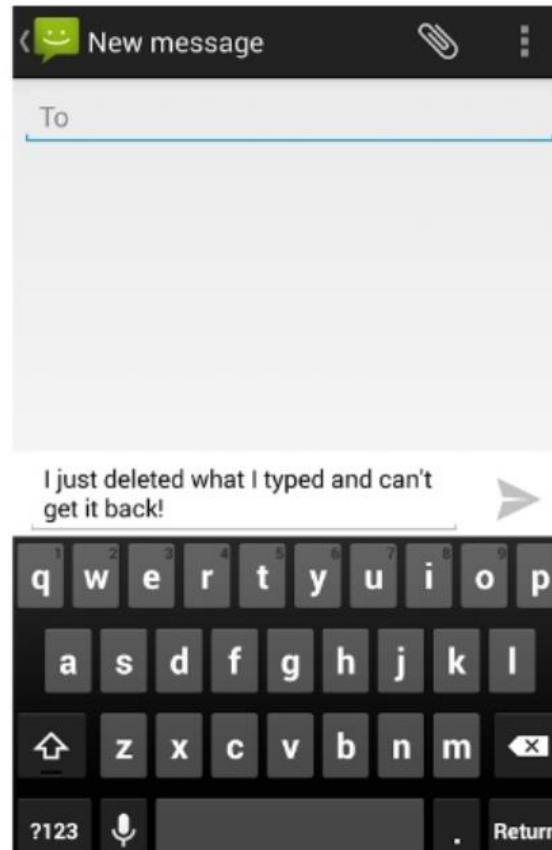


3. User control and freedom



There is no way to undo a text edit on Android!

Avoid pissing the user off by giving him no choice to revert to an earlier state.

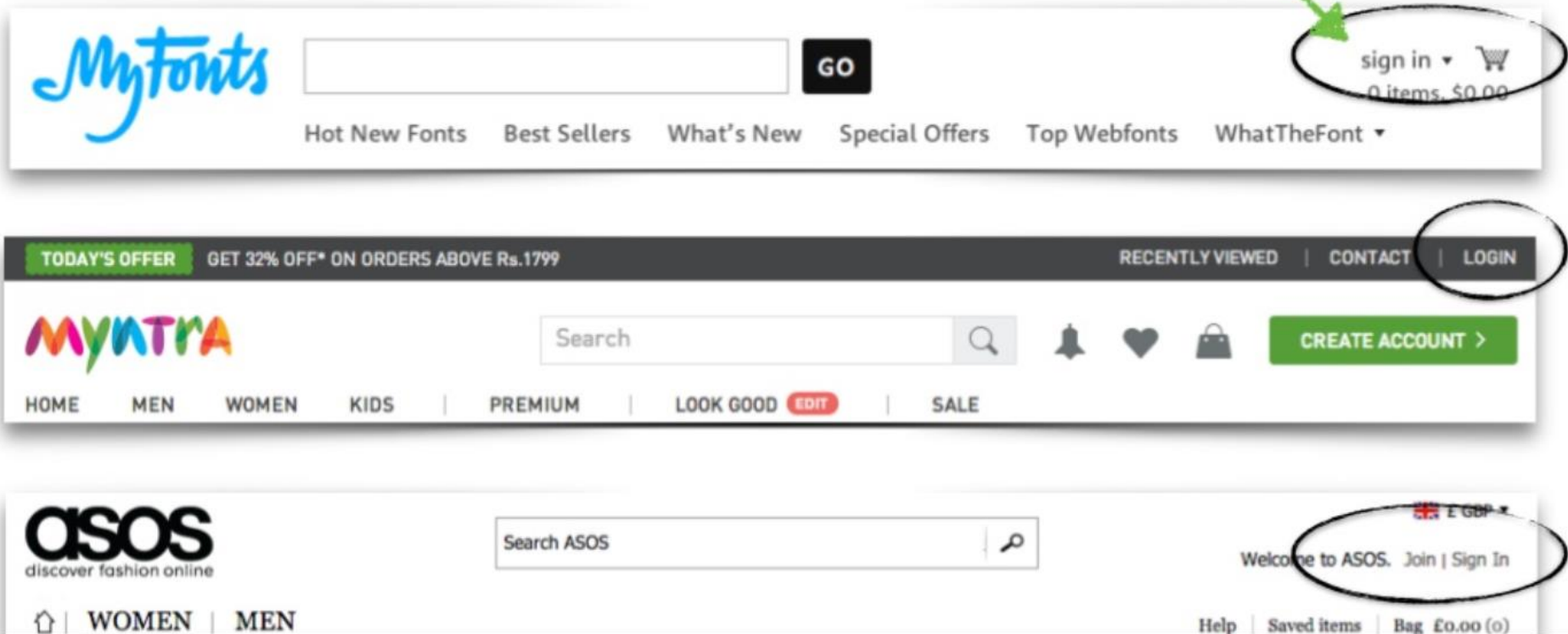


4. Consistency and standards

Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.

4. Consistency and standards

As a convention, people expect call to actions like Sign In to be at the top right on any website

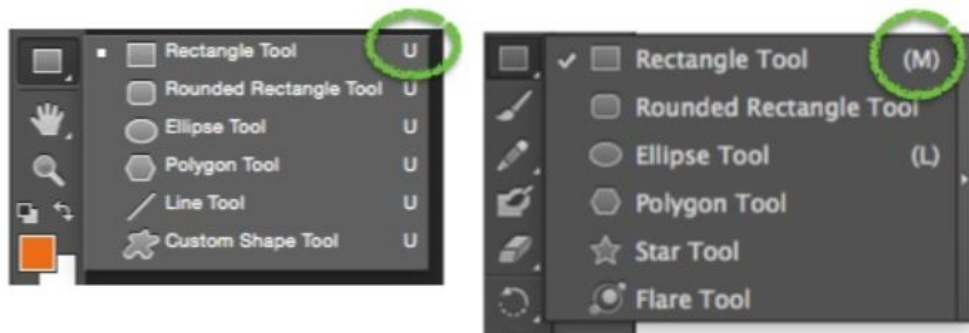


4. Consistency and standards



One would expect the knob to be the volume control, but it's not.

Similar functions have different shortcuts in Adobe's various software.



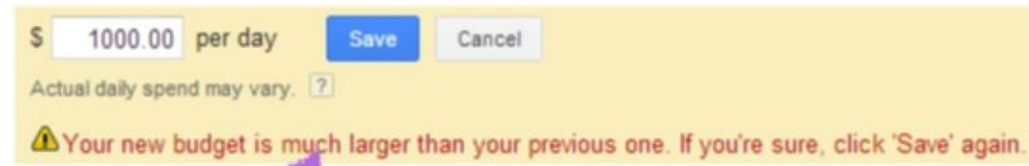
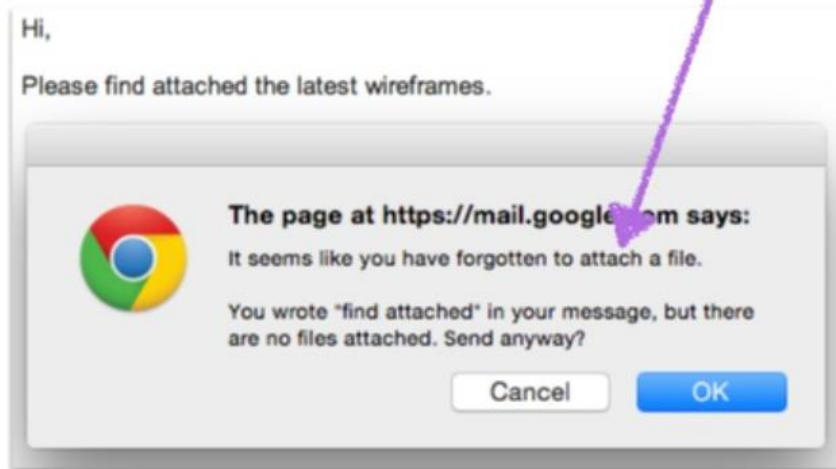
5. Error prevention

Even better than good error messages is a careful design which prevents a problem from occurring in the first place.

Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.

5. Error prevention

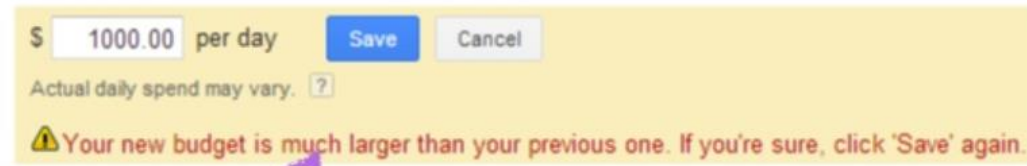
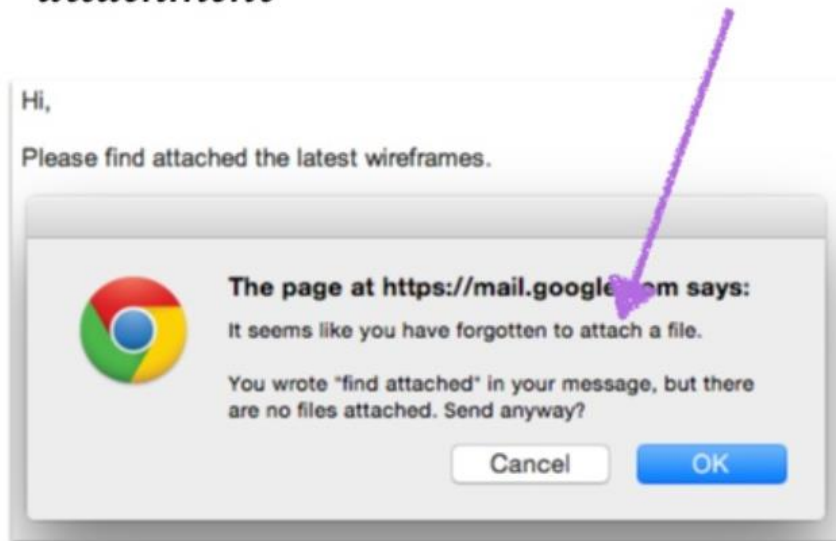
Gmail prompts you when you forget to insert an attachment



On Google Ads, if you input a budget much larger than your previous one, the system checks with you if you really meant to do so.

5. Error prevention

Gmail prompts you when you forget to insert an attachment



On Google Ads, if you input a budget much larger than your previous one, the system checks with you if you really meant to do so.

5. Error prevention



Facebook does not try and prevent you from posting an ad that is against its guidelines. e.g. There is no way for a first time user to assume that his ad might get pulled off mid-campaign because Facebook might later find it inappropriate.

Post	Total Reach?	Paid Reach?	Promotion
Thursday Facebook Special One Day O...	714	435	Error

Your Post wasn't boosted because it uses too much text in its image, which violates Facebook's ad guidelines. Sponsored images that show in the News Feed can't include more than 20% text. The post remains published, but it is not running as an ad. You'll still be charged for any impressions or clicks your ad received before it was disapproved.

Learn more or upload your ad image to see why it is considered 20% text. Get in touch if you need help.

435	19	\$4.07
Paid Reach [?]	Engagement [?]	Budget Spent

Boost your post to reach more people. [Learn more.](#)

Audience

- ☐ People who like your Page and their friends [?]
- ☒ People you choose through targeting [?]

Location

Cities + 10 miles

Bangalore, India ×
Mumbai, India × Pune, India ×

Age

20 - 31

WE'RE HIRING

6. Recognition rather than recall

Make objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.

6. Recognition rather than recall

Clear proposition

Search for hotels

Over 100,000 hotels around the world

☒ Hotels ☐ Weekend Getaways NEW

Where
Enter a city, area, landmark or hotel

Check-in
Pick a date

Check-out
Pick a date

Rooms
1

Adults
1
12+ yrs

Children
0
0-11 yrs

Search hotels

Escape Every Weekend

Instructions prompting the user to explore

A single bold call to action leaves no room for distraction

The image shows a screenshot of a travel website's search interface. On the left is a vertical sidebar with navigation links: Flights, Hotels, Flight + Hotel, Trains, Buses, Weekend getaways (marked with a 'NEW' badge), Collections, Waytogo, and Mobile. The main content area is titled 'Search for hotels' with the subtitle 'Over 100,000 hotels around the world'. Below this are two radio buttons for 'Hotels' (selected) and 'Weekend Getaways' (with a 'NEW' badge). A text input field labeled 'Where' prompts the user to 'Enter a city, area, landmark or hotel'. Below the input field are date pickers for 'Check-in' and 'Check-out', each labeled 'Pick a date'. Further down are dropdown menus for 'Rooms' (set to 1), 'Adults' (set to 1, with '12+ yrs' below), and 'Children' (set to 0, with '0-11 yrs' below). At the bottom right is a large orange button labeled 'Search hotels'. Annotations include a teal arrow pointing to the 'Search for hotels' title with the text 'Clear proposition'; a teal arrow pointing to the 'Weekend Getaways' radio button with the text 'Instructions prompting the user to explore' and a circular graphic containing a car icon and the text 'WEEKEND GETAWAYS' and 'Escape Every Weekend'; and a teal arrow pointing to the 'Search hotels' button with the text 'A single bold call to action leaves no room for distraction'.

6. Recognition rather than recall



Important actions should be easy to access and identify.

In an earlier version of Windows 8, it was almost impossible to shut down the computer without googling how to do it.

How To Shutdown Windows 8

1. Close all desktop apps you have opened.
2. Swipe in from the right edge of the screen, and then tap Settings. If you're using a mouse, point to the upper-right corner of the screen, move the mouse pointer down, and then click Settings.

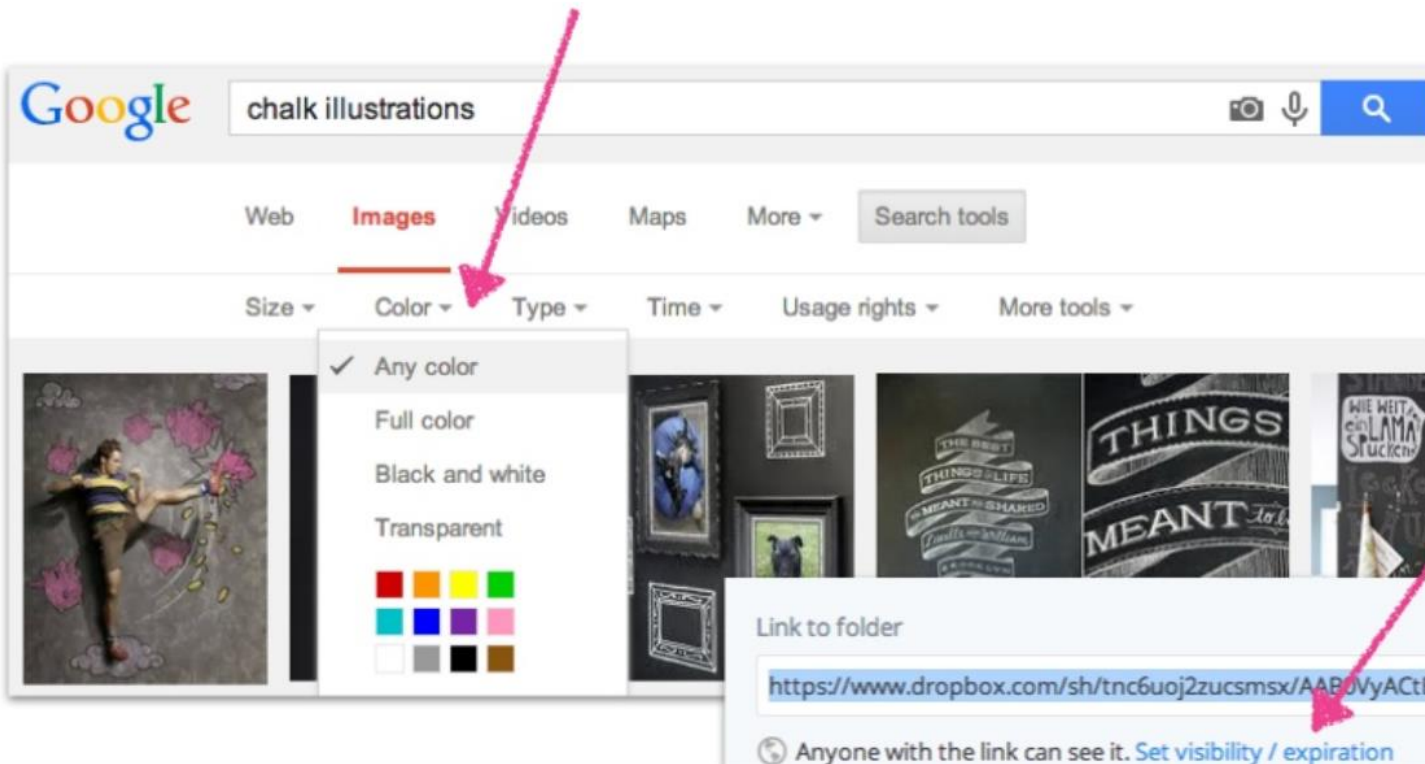


7. Flexibility and efficiency of use

Accelerators -- unseen by the novice user -- may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.

7. Flexibility and efficiency of use

Advanced users can use filters on Google Images to narrow down their search results.



When you share a file on Dropbox, you can simply copy the link, or set advanced options.

7. Flexibility and efficiency of use



This feature request highlights the frustration of an advanced user not being able to customise default features.

810
votes

Vote

Please allow custom labels for phone numbers, emails, addresses. home 1, home 2, work 1, work 2 just doesn't cut it any more.

Please allow custom labels for phone numbers, emails, addresses. home 1, home 2, work 1, work 2 just doesn't cut it any more.

I want to label multiple telephone numbers for a frequent traveler as "mobile (usa)" or "mobile (aus)" or mobile (eur). Or for my favorite pizza chain as downtown and midtown. Or I don't want to create my customer's secretary as a separate contact instead I want to put her under my customer's contact card and label her phone as "Cindy".

NEW PHONE CONTACT

photo

name

email

+ phone

+ email

+ ringtone

website

www.windowphonetools.com

+ other

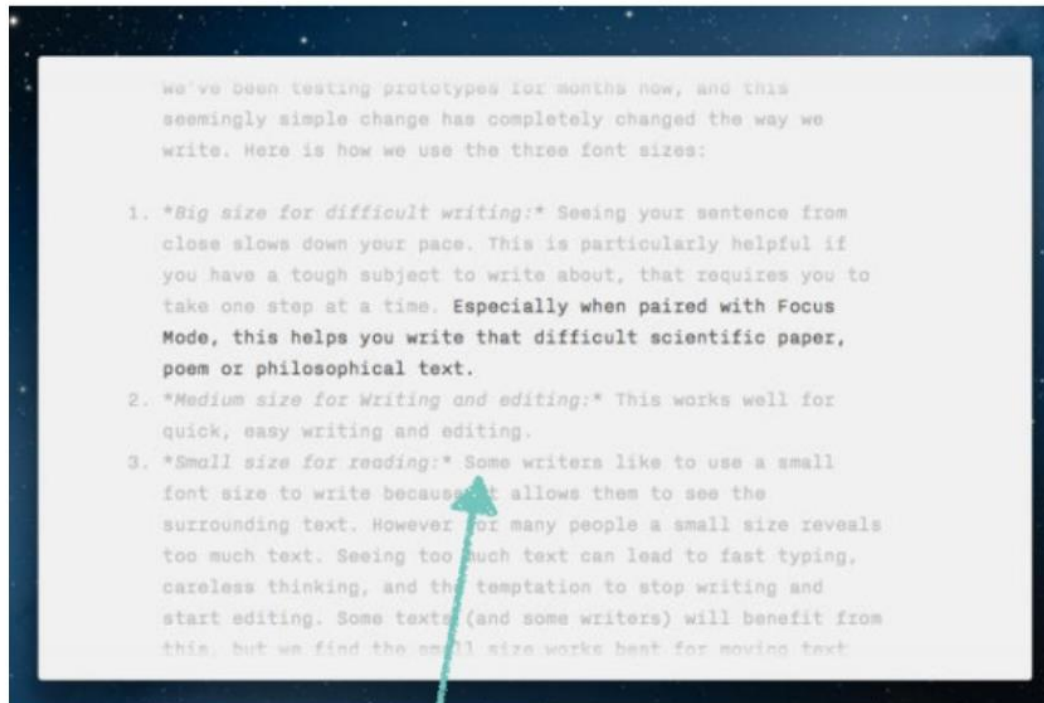
...

8. Aesthetic and minimalist design

Dialogues should not contain information which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.

8. Aesthetic and minimalist design

Dieter Ram's designs reflect this principle. "Less, but better – because it concentrates on the essential aspects, and the products are not burdened with non-essentials."

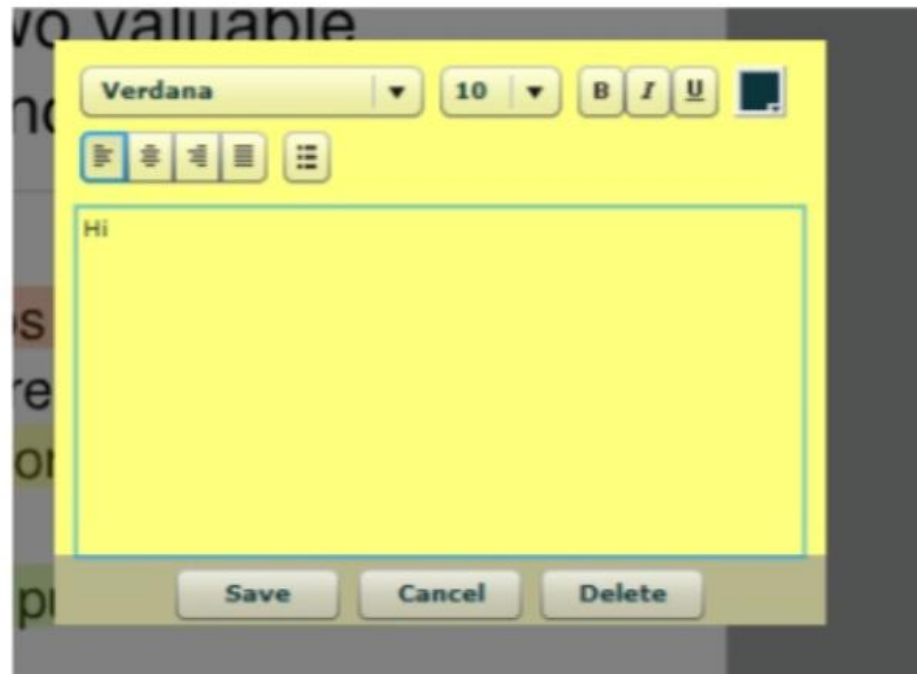


The iA Writer app is a clean typing sheet with no distractions in the interface. It allows you to focus on what you're writing and hides everything else.

8. Aesthetic and minimalist design



Avoid overloading the interface with features . Examine every element and ask - is this really needed?

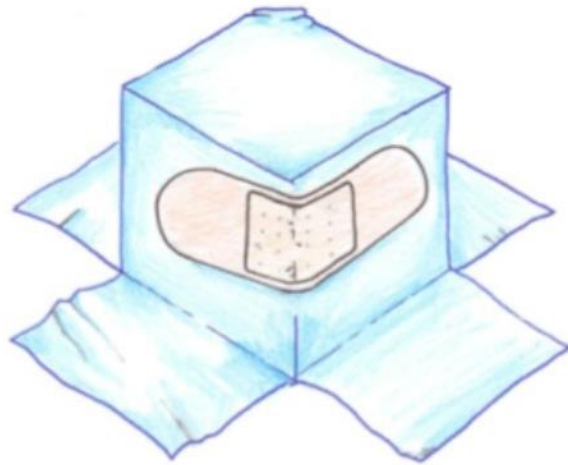


9. Help users recognize, diagnose, and recover from errors

Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

9. Help users recognize, diagnose, and recover from errors

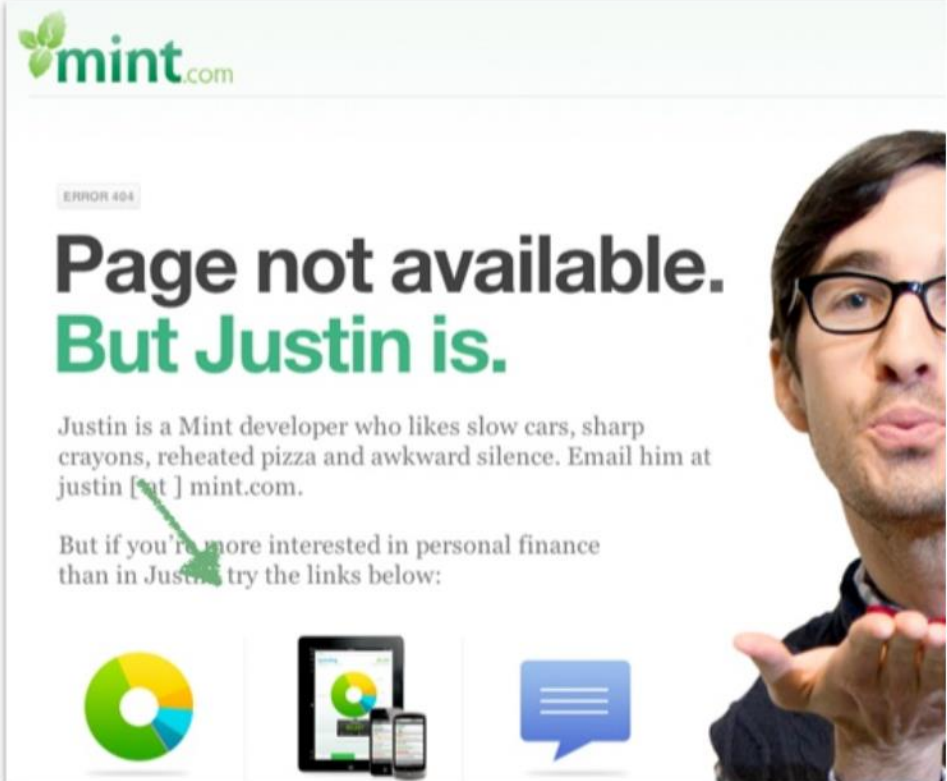
An assuring error message on Dropbox



Error

Something went wrong. Don't worry, your files are still safe and the Dropboxers have been notified. Check out our [Help Center](#) and [forums](#) for help, or head back to [home](#).

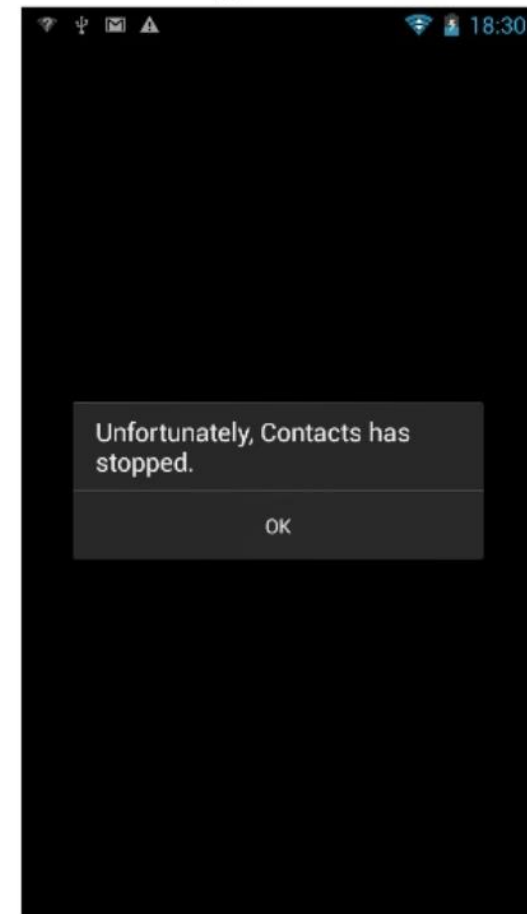
A funny message keeps the audience engaged, while relevant links make sure they stay on your website.



9. Help users recognize, diagnose, and recover from errors



Don't tell people that something's broken and can't be fixed.



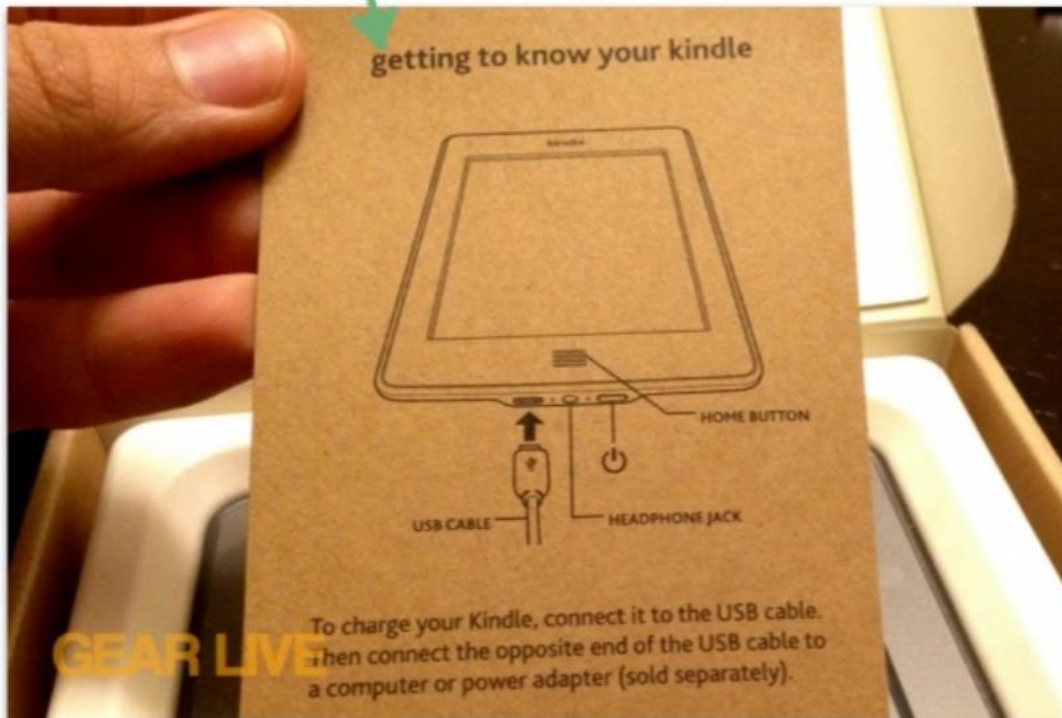
10. Help and documentation

Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation.

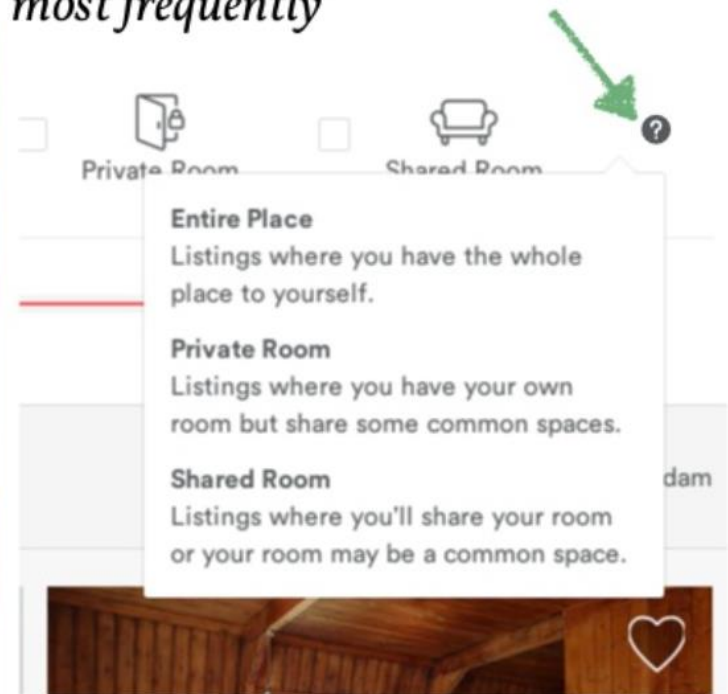
Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.

10. Help and documentation

The Kindle comes with an instruction card tucked inside the box flap, instead of a long boring manual



Contextual help on AirBnB provides explanations where they are needed most frequently



The background features a complex network of white lines connecting small white dots on an orange-to-red gradient. On the left, there are several diagonal stripes: a thick grey one, a thinner light blue one, and two orange ones of varying widths.

Web Design & Usability guidelines

https://www.usability.gov/sites/default/files/documents/guidelines_book.pdf

The *Guidelines* provide a clear sense of the range of issues that designers need to consider when planning and designing a Web site.

Within each chapter of this book, the guidelines are ordered according to their 'Relative Importance' ratings.

Some 'Strength of Evidence' ratings are low because there is a lack of research for that particular issue.



Research-Based Web Design & Usability Guidelines

Forewords by:

Michael O. Leavitt

Secretary of Health and Human Services

Ben Shneiderman

Professor of Computer Science, University of Maryland



5 – Strong Research Support 12345

- Cumulative and compelling, supporting research-based evidence
- At least one formal, rigorous study with contextual validity
- No known conflicting research-based findings
- Expert opinion agrees with the research

4 – Moderate Research Support 1234○

- Cumulative research-based evidence
- There may or may not be conflicting research-based findings
- Expert opinion
 - Tends to agree with the research, and
 - A consensus seems to be building

3 – Weak Research Support 123○○○

- Limited research-based evidence
- Conflicting research-based findings may exist
 - and/or -
- There is mixed agreement of expert opinions

2 – Strong Expert Opinion Support 12○○○

- No research-based evidence
- Experts tend to agree, although there may not be a consensus
- Multiple supporting expert opinions in textbooks, style guides, etc.
- Generally accepted as a 'best practice' or reflects 'state of practice'

1 – Weak Expert Opinion Support 1○○○○

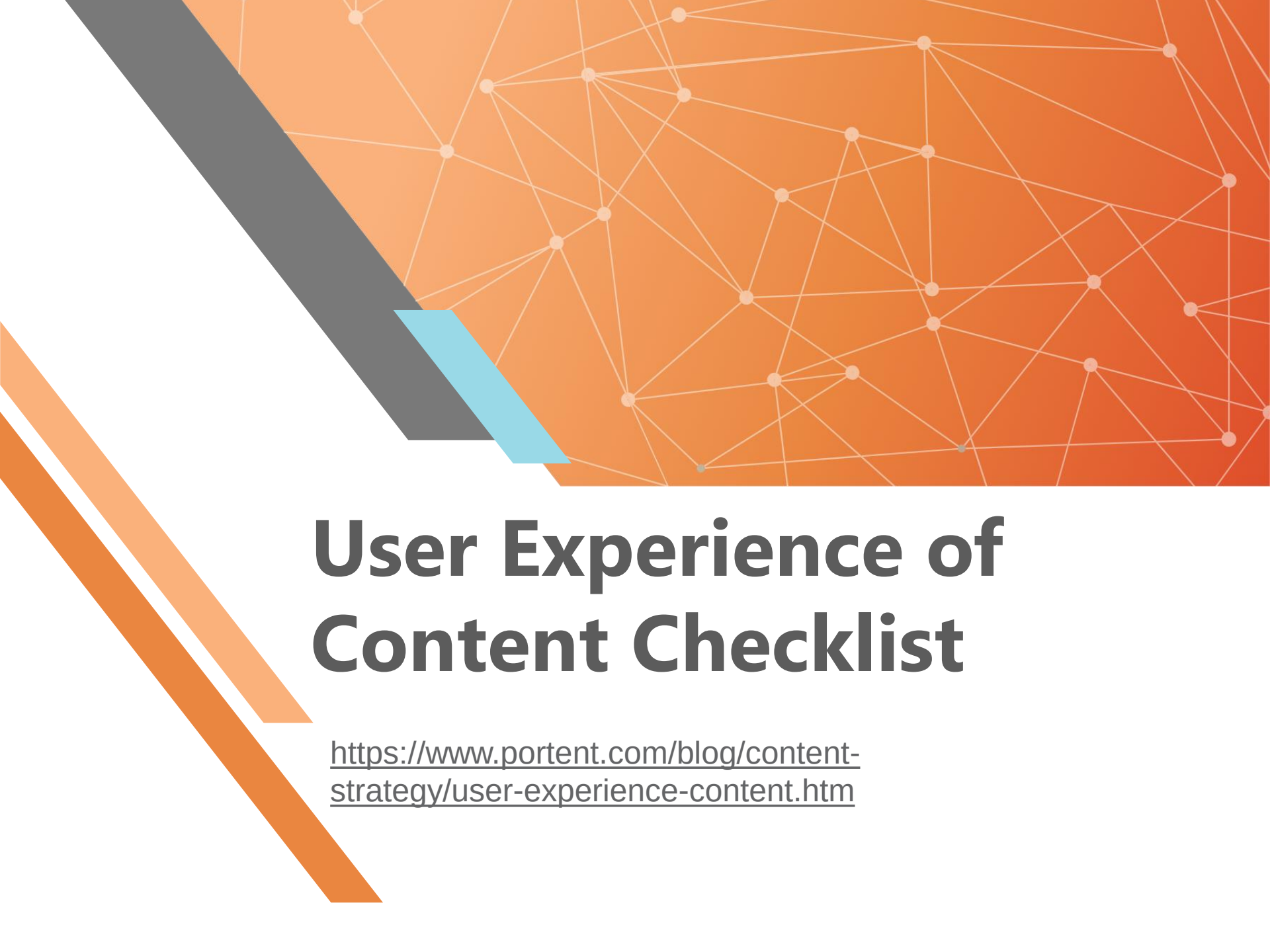
- No research-based evidence
- Limited or conflicting expert opinion

Chapters

- Chapter 1—Design Process and Evaluation
- Chapter 2—Optimizing the User Experience
- Chapter 3—Accessibility
- Chapter 4—Hardware and Software
- Chapter 5—The Homepage
- Chapter 6—Page Layout
- Chapter 7—Navigation
- Chapter 8—Scrolling and Paging
- Chapter 9—Headings, Titles, and Labels
- Chapter 10—Links
- Chapter 11—Text Appearance
- Chapter 12—Lists
- Chapter 13—Screen-Based Controls (Widgets)
- Chapter 14—Graphics, Images, and Multimedia
- Chapter 15—Writing Web Content
- Chapter 16—Content Organization
- Chapter 17—Search
- Chapter 18—Usability Testing

Guidelines Ranked by Relative Importance

Chapter: Guideline #	Guideline Heading	Relative Importance
1:1	Provide Useful Content	5
1:2	Establish User Requirements	5
1:3	Understand and Meet User's Expectations	5
1:4	Involve Users in Establishing User Requirements	5
2:1	Do Not Display Unsolicited Windows or Graphics	5
3:1	Comply with Section 508	5
3:2	Design Forms for Users Using Assistive Technology	5
3:3	Do Not Use Color Alone to Convey Information	5
5:1	Enable Access to the Homepage	5
5:2	Show All Major Options on the Homepage	5
5:3	Create a Positive First Impression of Your Site	5
6:1	Avoid Cluttered Displays	5
6:2	Place Important Items Consistently	5
6:3	Place Important Items at Top Center	5
8:1	Eliminate Horizontal Scrolling	5
9:1	Use Clear Category Labels	5
10:1	Use Meaningful Link Labels	5
13:1	Distinguish Required and Optional Data Entry Fields	5
13:2	Label Pushbuttons Clearly	5
15:1	Make Action Sequences Clear	5
16:1	Organize Information Clearly	5
16:2	Facilitate Scanning	5
16:3	Ensure that Necessary Information is Displayed	5
17:1	Ensure Usable Search Results	5
17:2	Design Search Engines to Search the Entire Site	5
1:5	Set and State Goals	4

The background features a network diagram of white dots connected by thin white lines on an orange gradient. On the left, there are several diagonal stripes: a thick grey one, a thinner light blue one, and two orange ones of varying shades.

User Experience of Content Checklist

<https://www.portent.com/blog/content-strategy/user-experience-content.htm>

113 Design Guidelines for Homepage Usability

- <https://www.nngroup.com/articles/113-design-guidelines-homepage-usability/>

NN/g Nielsen Norman Group

Evidence-Based User Experience Research, Training, and Consulting

Search

[HOME](#) [TRAINING & EVENTS](#) [CONSULTING](#) [REPORTS](#) [ARTICLES](#) [ABOUT NN/G](#)

Topics

[E-commerce](#)

[Intranets](#)

113 Design Guidelines for Homepage Usability

by **JAKOB NIELSEN** on October 31, 2001

Topics: [Web Usability](#)

247 web usability guidelines

- <https://www.userfocus.co.uk/resources/guidelines.html>

Web usability guidelines

- [Home page usability](#): 20 guidelines to evaluate the usability of home pages.
- [Task orientation](#): 44 guidelines to evaluate how well a web site supports the users tasks.
- [Navigation and IA](#): 29 guidelines to evaluate navigation and information architecture.
- [Forms and data entry](#): 23 guidelines to evaluate forms and data entry.
- [Trust and credibility](#): 13 guidelines to evaluate trust and credibility.
- [Writing and content quality](#): 23 guidelines to evaluate writing and content quality.
- [Page layout and visual design](#): 38 guidelines to evaluate page layout and visual design.
- [Search usability](#): 20 guidelines to evaluate search.
- [Help, feedback and error tolerance](#): 37 guidelines to evaluate help, feedback and error tolerance.



[Download an Excel workbook containing all 247 web usability guidelines](#)



Designing the UX

What is a UX Designer?

- A UX designer is someone who investigates and analyzes how users feel about the products he or she offers them.
- UX designers then apply this knowledge to product development in order to ensure that the user has the best possible experience with a product.
- UX designers conduct research, analyze their findings, inform other members of the development team of their findings, monitor development projects to ensure those findings are implemented, and do much more.

T-shirt of an UX designer

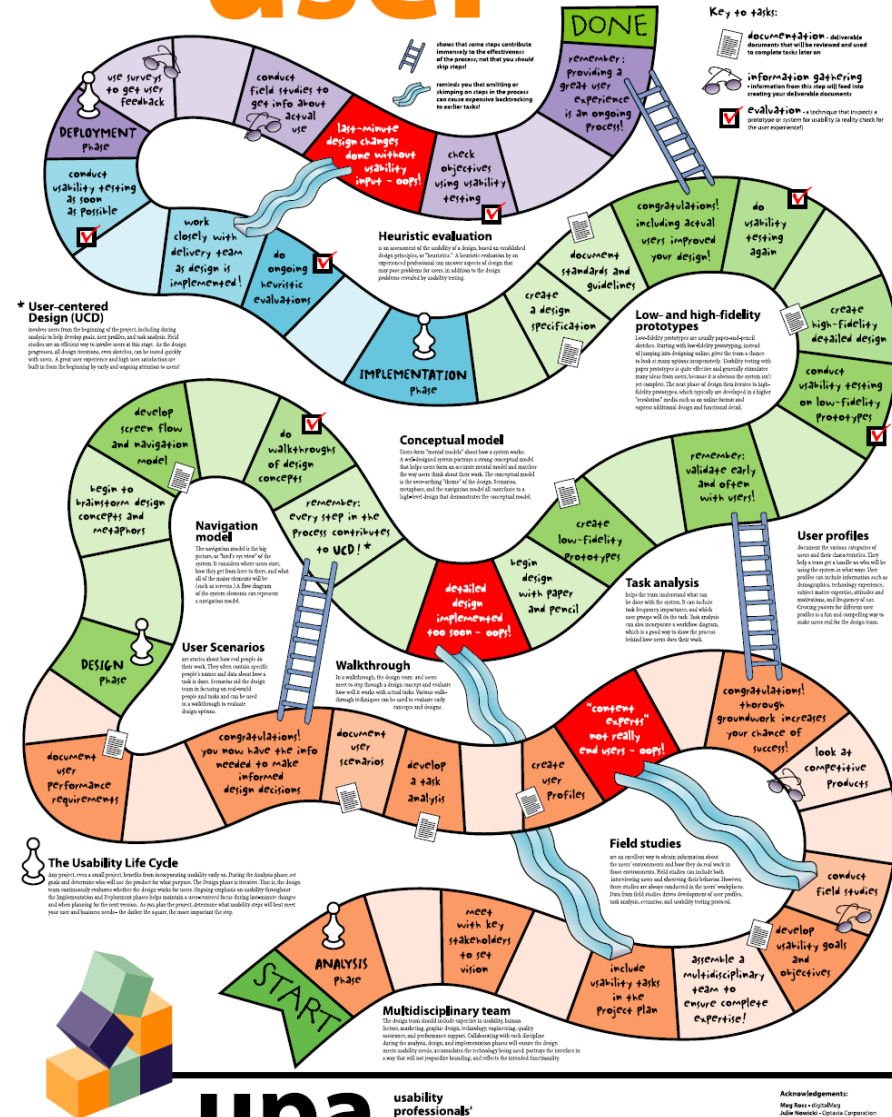


“It Depends...”

- The term ‘it depends’ is often used when asked if a product is ‘usable’ or a ‘good user experience’
- Again, all experiences are contextual, so often the same product can be ‘usable’ for one person and ‘not usable’ for another. This is one of the more difficult aspects of user experience is to understand
- User research helps determine what aspects of context are most important for a good user experience

Analysis Phase

- Meet with key stakeholders to set vision
- Include usability tasks in the project plan
- Assemble a multidisciplinary team to ensure complete expertise
- Develop usability goals and objectives
- Conduct field studies
- Look at competitive products
- Create user profiles
- Develop a task analysis
- Document user scenarios
- Document user performance requirements



Designing the UX

Design Phase

Begin to brainstorm design concepts and metaphors

Develop screen flow and navigation model

Do walkthroughs of design concepts

Begin design with paper and pencil

Create low-fidelity prototypes

Conduct usability testing on low-fidelity prototypes

Create high-fidelity detailed design

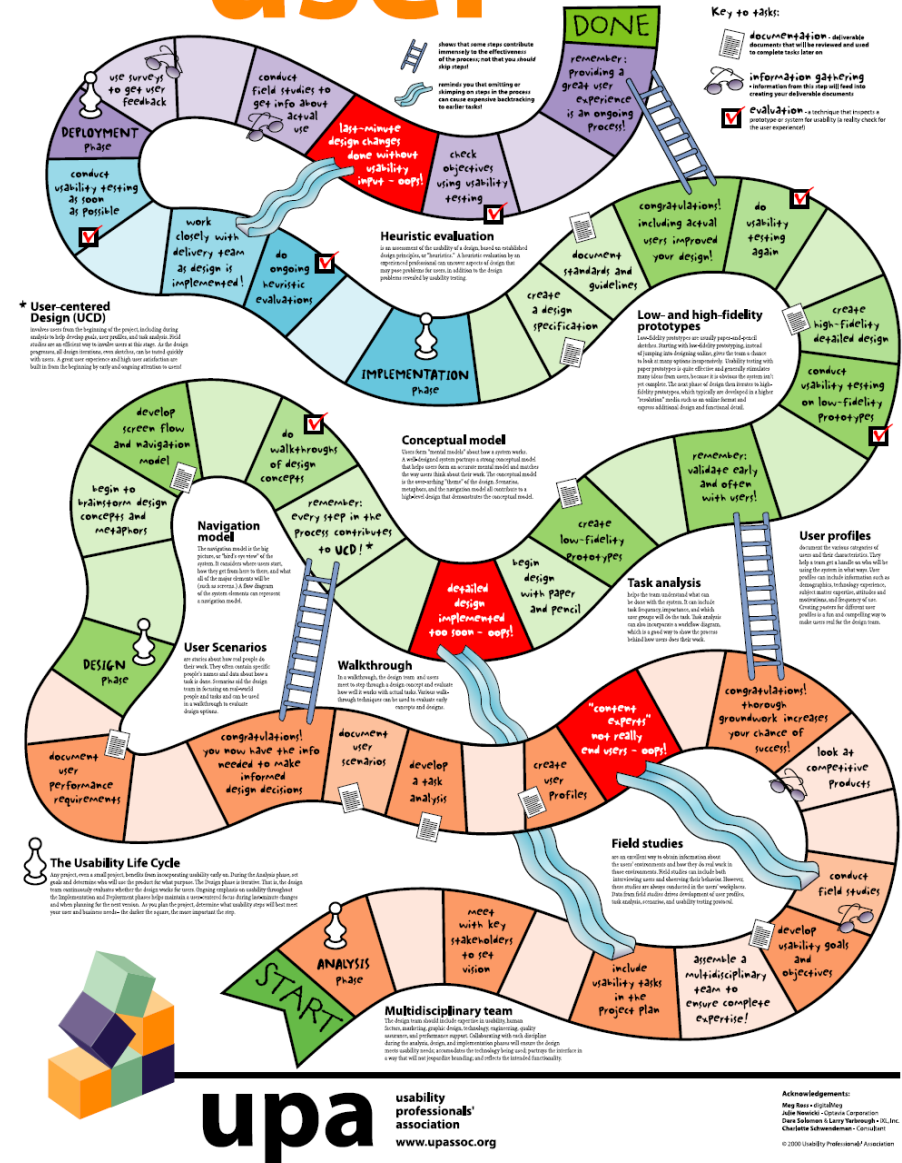
Do usability testing again

Document standards and guidelines

Create a design specification

designing the **user** experience

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Charlotte Schwenk - Consultant
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Designing the UX

Implementation Phase

Do ongoing heuristic evaluations

Work closely with delivery team as design is implemented

Conduct usability testing as soon as possible

Deployment Phase

Use surveys to get user feedback

Conduct field studies to get info about actual use

Check objectives using usability testing

designing the **user** experience

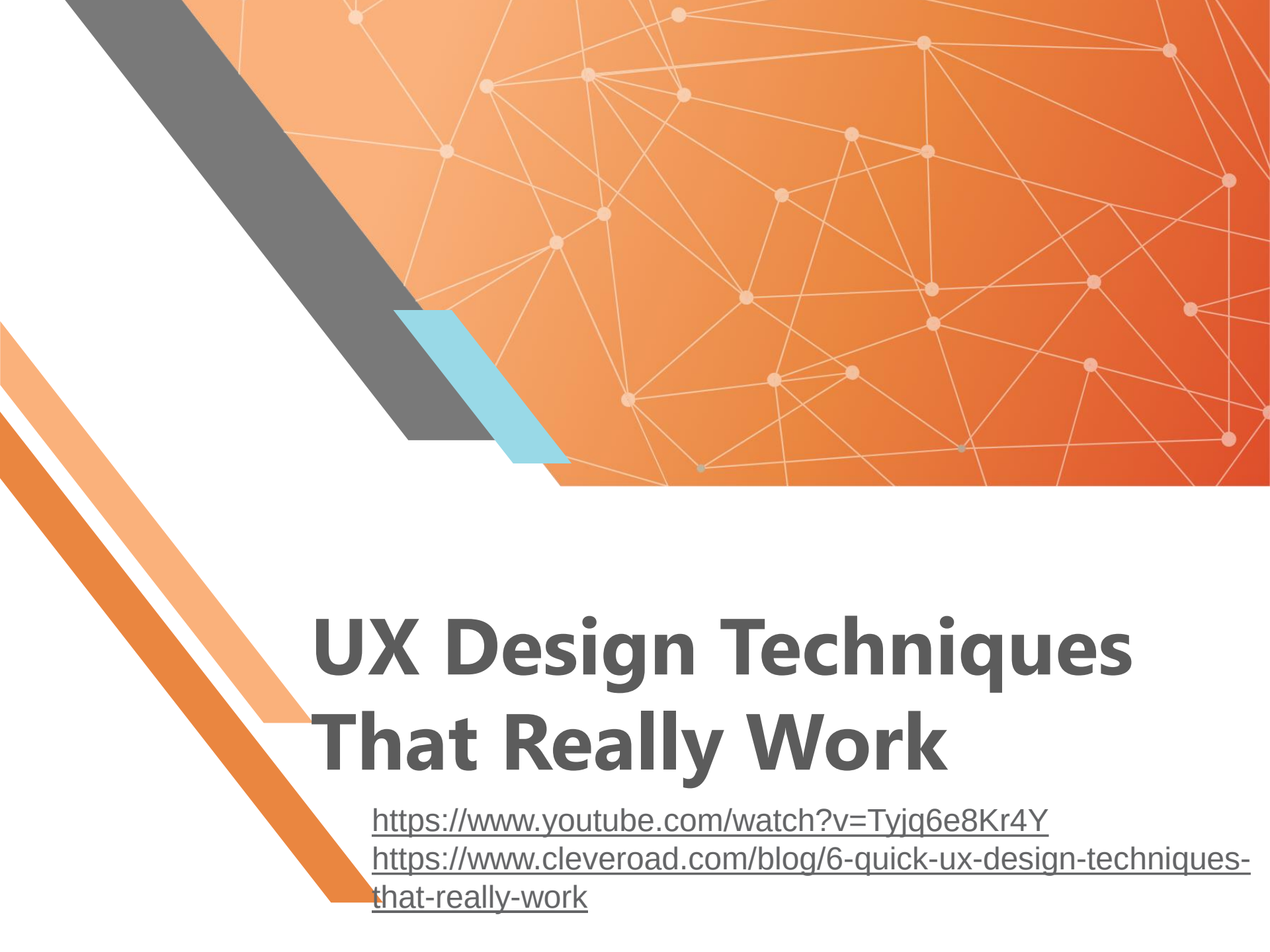
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UX Design Techniques That Really Work

<https://www.youtube.com/watch?v=Tyjq6e8Kr4Y>

<https://www.cleveroad.com/blog/6-quick-ux-design-techniques-that-really-work>

Personas

- First of all, you need to identify people who are going to use your website/app. (Learn a target audience of your product.)
- For that, you create fictional characters- personas.
- Personas are composed on the basis of the data that you receive during interviewing, testing, and analyzing typical users.
- The main purpose of making them is to define the most common needs, expectations, and limitations of your customers.

Personas

Effective personas:

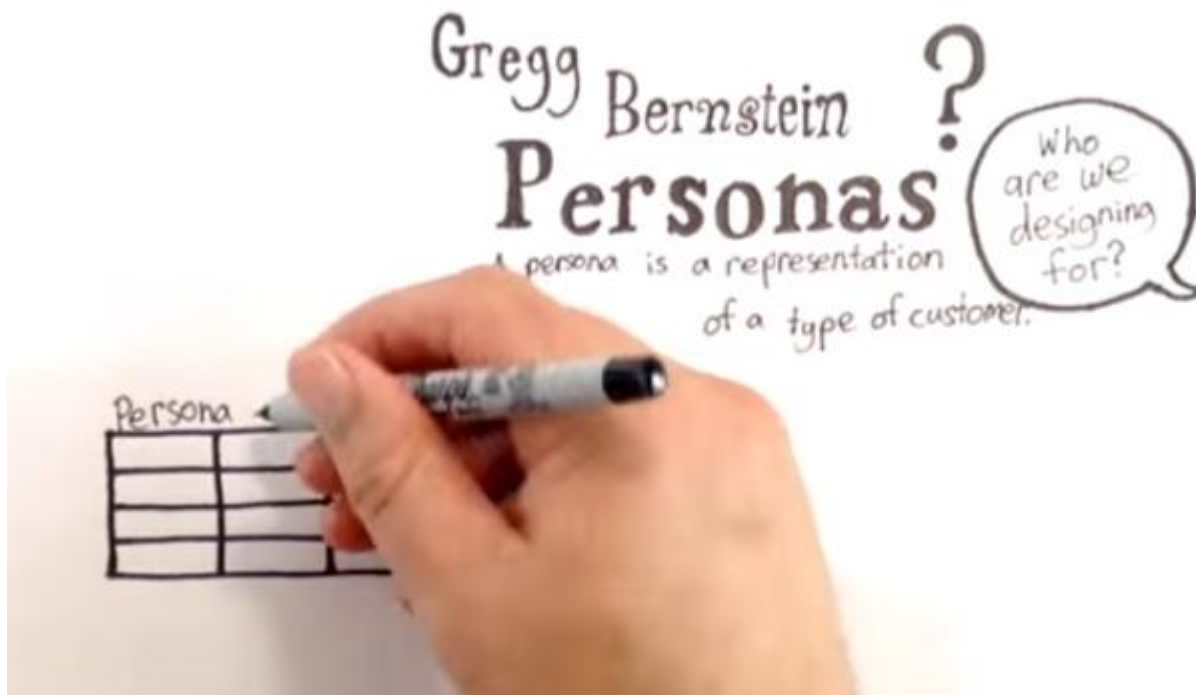
- Represent a major user group for your website
- Express and focus on the major needs and expectations of the most important user groups
- Give a clear picture of the user's expectations and how they're likely to use the site
- Aid in uncovering universal features and functionality
- Describe real people with backgrounds, goals, and values

Benefits of Personas

- Stakeholders and leaders evaluate new site feature ideas
- Information architects develop informed wireframes, interface behaviors, and labeling
- Designers create the overall look and feel of the website
- System engineers/developers decide which approaches to take based on user behaviors
- Copy writers ensure site content is written to the appropriate audiences

Personas

- <https://www.youtube.com/watch?v=B23iWg0koi8>



Personas

ALISON



BUSY MOM

About Alison

- Alison is a middle aged woman who lives in San-Francisco and works for an audit firm;
- She is married and has 2 young kids;
- Alison has a busy life so she prefers online shopping which greatly saves her time;
- She simply wants to find the needed products as soon as possible;
- She works hard, so she isn't going to overpay and seeks to achieve the quality that meets the price;
- Alison gets annoyed when the app displays results that do not meet her requirements;
- She feels frustrated when the app works slowly or crashes at all.

User stories, scenarios, and storyboards

- When finished with personas, you may proceed with user stories, scenarios, and storyboards.
- They describe particular cases of interacting with your app.
- The difference is that the
 - first one gives a general description,
 - the second - a more detailed one,
 - the third - a visual demonstration.

User stories, scenarios, and storyboards

- **User story or epic** gives a rough idea about a particular action or set of actions.
- E.g. "Alison orders a teddy bear to the birthday of her kid".

Scenario

Now, it's time to expand on the topic. Here is how the relevant **scenario** is going to look:

1. It's Sunday morning and Alison has some spare time to choose a present for her child.
2. She takes her smartphone and starts "Micky", online shopping app that specializes in baby stuff.
3. She knows exactly what she wants to buy, so she selects *Browse by category -> Toys -> Fluffy Toys -> Stuffed Animals -> Teddy Bears*.
4. She ticks the boxes size - 27in, condition - new.
5. Having found an appropriate bear, she clicks *"Add to the shopping cart"*.
6. After that, she opens shopping cart, clicks *"Buy"*, and enters her cc credentials.
7. She accepts terms and conditions and confirms the order by clicking *"Confirm order"* button.
8. In few minutes she receives a confirmation email with the order number and expected date of delivery.

How to Write User Story Scenario?



<https://www.youtube.com/watch?v=4lvijAfjb4U&pbjreload=10>

Storyboard

Let's see how the story is going to be illustrated with the help of **storyboard**:



STEP 1

Browse by category

Clothing
Shoes
Accessories
Toys → Fluffy Toys → Stuffed Animals
→ Teddy Bears

Alison clicks "Browse by category"



STEP 2

Tick boxes

Gender: Boys, Boys&Girls, Girls, Not specified
Brand: Disney, LEGO, Chicco, Unbranded
Size: 16In, 22In, 24In, 25In, 27In, 30In
Condition: New, Used, Not Specified

Alison ticks the boxes size - 27In, condition - new.



STEP 3

Payment options

Invoice
Credit card
Cheque

Alison opens shopping cart, click "Buy", and chooses payment option.



STEP 4

Credit card credentials

Card number
Expiration date
Security code

Alison enters cc credentials.



STEP 5

Terms and conditions

Terms and conditions text

Alison views the terms and conditions



STEP 6

Agreement

I accept the terms and conditions

Alison accepts the terms and conditions



STEP 7

Summary

You have successfully placed an order #123. Check your email to see a detailed information about the order.

Alison sees that the order was placed successfully.



STEP 8

Feedback

Baby Store "Mickey"
To: Alison Williams
Order #123

Dear Alison! Thank you for choosing The Micky store. Your order #123 has been successfully placed...

Alison receives a confirmation email with the order number and expected date of delivery.

Writing-first design technique

- Unless you are a newcomer in design area, you would not start designing with special tools like Photoshop or 3D max at once.
- Before using any software or online program, professional designers make a sketch of the application manually, and before drawing, they usually use a worded description.

Writing-first design technique

- It will help you systematize the information.

To succeed, ask yourself a few questions, like:

- What is special about the client?
- What is the purpose of their product?
- Which problems do they seek to solve?
- How can the ideas be brought to users?
- How can you make them interested and involved?

Writing-first design technique

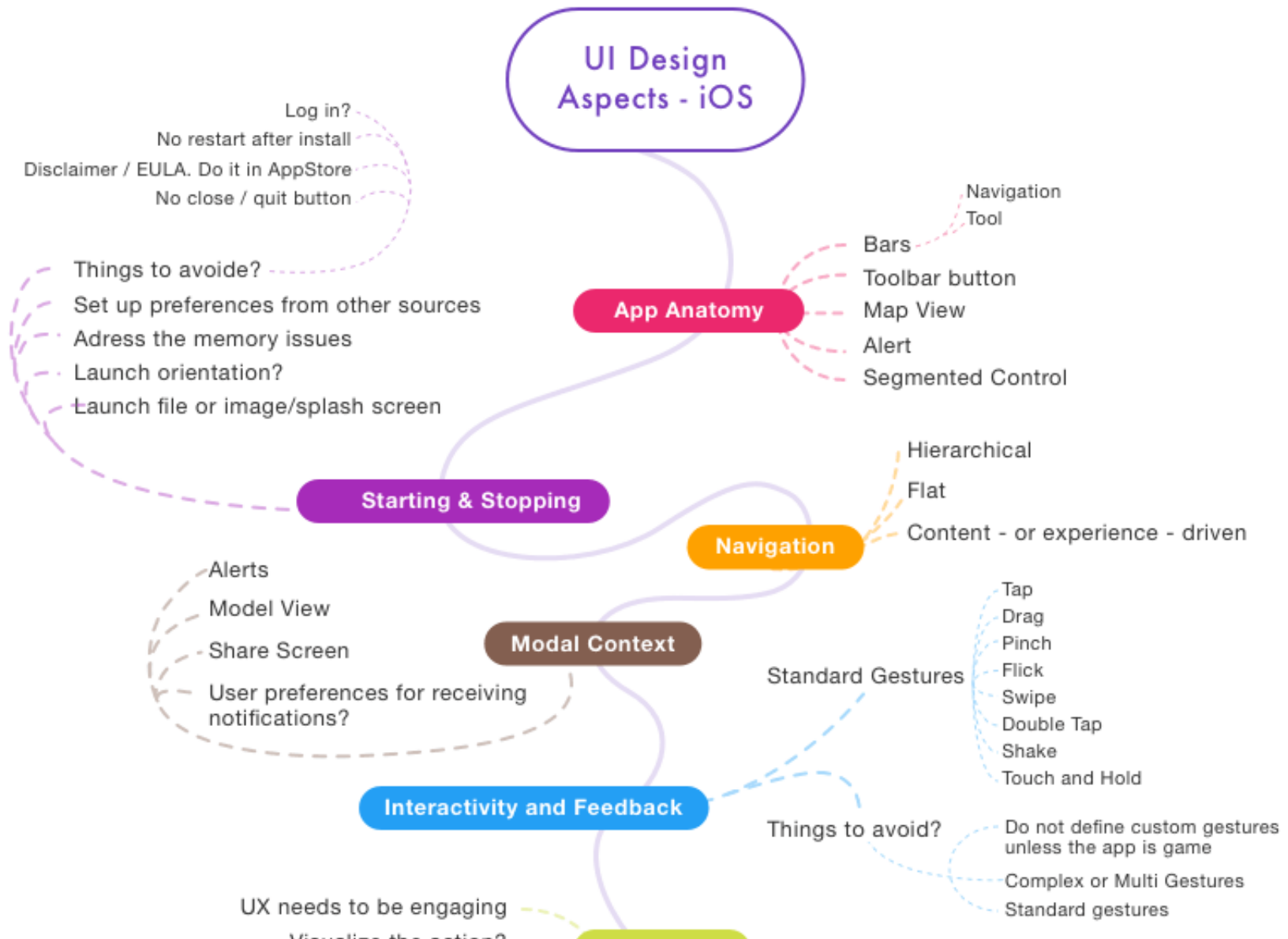
Make the notes by answering the following questions: Are the elements arranged logically?

- Can you find them fast and easy?
- Is the text placed correctly?
- Does it have a clear and contemporary style?
- Is it of the right length?
- Can you separate it to the different pages without losing the main idea?
- What else can you do to improve the existed design?

Mind-mapping

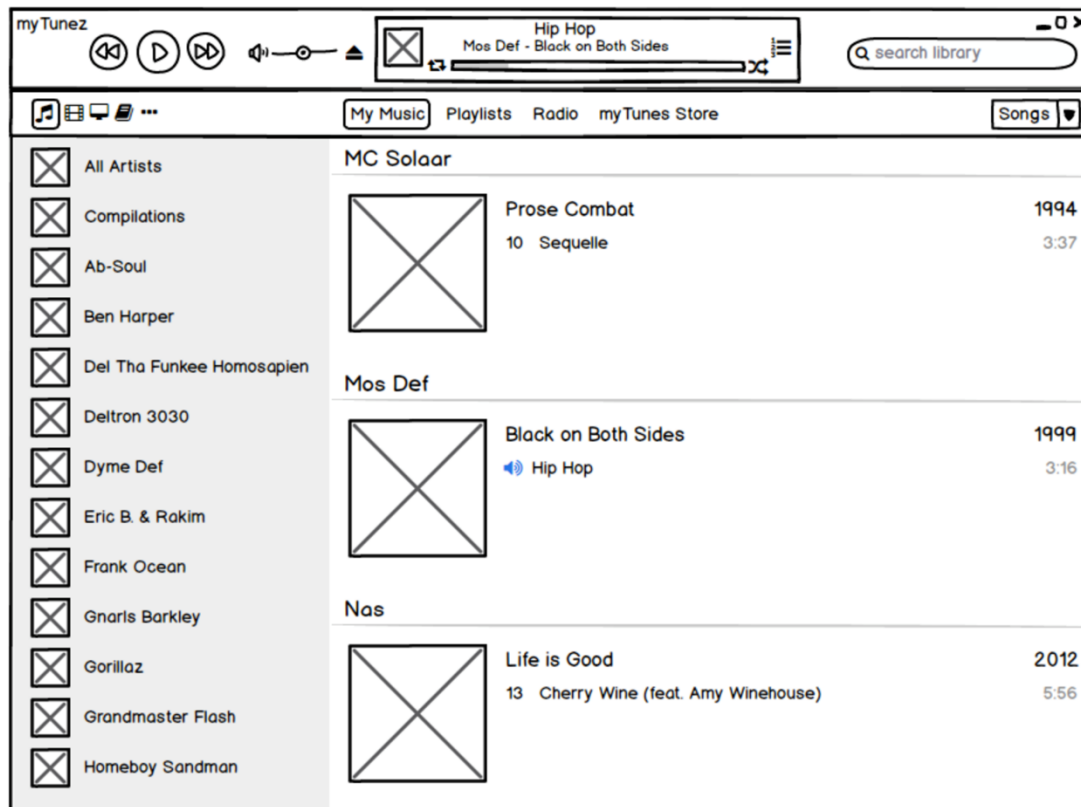
- It helps arrange the gathered information and introduce it as a visual output.
- Mind-mapping comes in handy when you want to set all the creative thoughts in one place and define the scheme of the app layout.

Mind-mapping



Wireframing and paper prototyping

A wireframe is the first and the roughest sketch of your product. It shows the placement of different components and their interrelation.



Paper prototype

- **Paper Prototyping** is a technique that consists of creating hand drawings of user interfaces in order to enable them to be rapidly designed, simulated and tested.
- The different screens are sketched prior to the testing.
- Typical use cases are then created and the users are requested to try and perform them by interacting with the prototype.
- Based on where they press or select, the human computer will change or alter the prototype by introducing whole screens or snippets, depending on how the system is intended to react.

Paper prototype testing

- Low fidelity prototype testing of the youtube website
- <https://www.youtube.com/watch?v=aDqbMOI13o4>



User testing

- User testing is the best approach to finding out whether you are on the right way to effective UX.
- The number of participants and the length of tests performing will depend on the project specific and budget size. Also, you can use moderated testing, unmoderated testing, or both.

Moderated testing

Moderated testing involves your presence and coordination of the users' actions.

The advantages of moderate testing are that:

- You watch users' reaction, act habit, and emotions.
- You track all the results in a real-time mode.
- You control all the steps and target them to the right point.

Unmoderated testing

Unmoderated testing means that the app is used without your participation. You may provide users with some test cases, however, this method usually implies no intervention.

The benefits of unmoderated testing are that:

- You save time and money since you do not need to participate personally.
- You may test features which you haven't even thought about.
- You receive the test results that come from a natural environment.

Card Sorting

- Card sorting is a method used to help design or evaluate the information architecture of a site.
- In a card sorting session, participants organize topics into categories that make sense to them and they may also help you label these groups.
- To conduct a card sort, you can use actual cards, pieces of paper, or one of several online card-sorting software tools.

Benefits of Card Sorting

Card sorting will help you understand your users' expectations and understanding of your topics.

Knowing how your users group information can help you:

- Build the structure for your website
- Decide what to put on the homepage
- Label categories and navigation

Card sorting in action

- <https://youtu.be/FTzHeYPB9c8?t=2m36s>



Eye tracking

- In the simplest terms, eye tracking is the measurement of eye activity.
 - Where do we look?
 - What do we ignore?
 - When do we blink?
 - How does the pupil react to different stimuli?

<http://www.eyetracking.com/>

Eye tracking

https://www.youtube.com/watch?v=lo_a2cfBUGc

Eye tracking



Heatmaps from user eyetracking studies of three websites. The areas where users looked the most are colored red; the yellow areas indicate fewer views, followed by the least-viewed blue areas. Gray areas didn't attract any fixations.

<https://www.nngroup.com/articles/f-shaped-pattern-reading-web-content-discovered/>

Eye tracking

Implications of the F Pattern

- **Users won't read your text thoroughly** in a word-by-word manner. Exhaustive reading is rare, especially when prospective customers are conducting their initial research to compile a shortlist of vendors. Yes, some people will read more, but most won't.
- **The first two paragraphs must state the most important information.** There's some hope that users will actually read this material, though they'll probably read more of the first paragraph than the second.
- **Start subheads, paragraphs, and bullet points with information-carrying words** that users will notice when scanning down the left side of your content in the final stem of their F-behavior. They'll read the third word on a line much less often than the first two words.



UX checklist

UX checklist

- <http://uxchecklist.github.io/>

Research Plan Explore	 Competitive Analysis See how others solve similar problems and try to not reinvent the wheel. Read more	 Data analysis Do you have all the useful data you need? Try to have a look at funnels, clicks, page views, performances... Read more	 User feedback Always speak with Customer Care team! Don't have one? Check your old surveys or videos, what your customer says? What do they actually do? Read more
	 User stories Have you done personas yet. If not DO IT NOW. Ok, now use them to write down user stories and scenarios. Read more	 User flows Create your user's flow based on the scenarios you created, you can use it later to review the journey and create wireframes on top of each step. Read more	 Red routes Define red routes for your product and you'll be able to identify, prioritise and eliminate any usability obstacles on key user journeys. Read more
	 Brainstorm & sketch Find a war room, fill it with markers and drinks, get together and sketch, discuss, vote, disrupt, have fun! Read more	 Wireframe Add some details and structure to your ideas, reuse patterns and create pages on top of your user flows so you'll not leave anything behind. Read more	 Prototype You can start creating paper prototypes and continuously iterate to more functional ones. Use sketches, HTML pages or static images, then just get some people and test. Read more

UX checklist

- <http://uxchecklist.github.io/>

Communicate Create Give feedback	 IA Understand your users, your data structure and your channels. How can you organise your navigation and content in a clear and consistent way? Read more	 Language Follow your brand personality, keep in mind users' culture and language, the context of your product and make sure they understand you. Read more	 Accessibility You don't need to add extra functionality or to duplicate any content. The key is simply to assess the requirements of those with different skills and limited devices. Read more
	 UI elements Reuse elements and patterns, follow your style guidelines, don't have one? Create your guidelines. Start small, then create pages. Read more	 Gestures So you have a swipe slider? Tell me more about pinch, drag, zoom, rotate, shake, six-inch smartphones, left handed people, mouseover, kinect, motion detection... Read more	 Responsiveness Can I see it on my mobile? Oh wait, what about my smart-watch which work as a remote for my 50" TV. Bonus: remember cross device experience. Read more
	 Waiting times If your users have to wait ages for the page to load, at least show them a loader, if take longer why don't you try something more entertaining? Read more	 Errors Be clear and specific on what and where user's error is. I mean, your error, because if it's your fault you should say it. Read more	 Completed actions Give immediate and clear feedback of successful user's actions. Do not always wait for server response, trust your server once in a while! Read more

UX checklist

- <http://uxchecklist.github.io/>

Analyse Delight Finalise	 Finalise layout It's time to let your design shine, make it in the right way, don't stop with the first solution, always ask "is this the best you can do?" Read more	 Use of images and icons Use of icons and images is strongly influenced by context, culture and layout that you use. Like icons, test your images, small changes can bring huge improvements. Read more	 Font & colours hierarchy Use colours and font sizes properly, try to follow your guidelines and keep it simple. The best visual hierarchies lead users to take the action confidently. Read more
	 Micro copy Every word is important, and a bit of personality will help your brand. Read more	 Micro interactions Trigger, rules, feedback, loop. Details make the product. Bonus: Ever heard about easter eggs? Read more	 Transitions Motion shouldn't be only beautiful, it should build meaning about the spatial relationships, functionality, and intention of the system. Read more
	 KPI Setup What you want to achieve? What are your goals? Write down how you define success and failure and check if you have everything you need to collect the data. Read more	 AB Test plan Plan your AB test ahead and, if you can, plan a short roadmap of improvements. Your goal is not just improving KPIs, but learning something. Read more	 Test UX lab, survey, sessions recording... test, observe and fix, test, observe and fix... Read more

Sources

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- <https://blog.prototypr.io/10-usability-heuristics-with-examples-4a81ada920c>
- Presentation of Christopher S. LaRoche. NEASIST User Experience & You: UX Design & Research January 6, 2016
- <https://www.usertesting.com/blog/2016/04/27/ui-vs-ux/>
- The Basics of User Experience Design BY INTERACTION DESIGN FOUNDATION



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Thank you for your attention

Web-development I. course